

Supplementary Material: Measuring Metric and Sampling Distortions in Penalty-Encoded QAOA Portfolio Benchmarks

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S-I. THE LITERATURE CENSUS

Sec. II summarises what fourteen published QAOA and annealing portfolio benchmarks report. The claim is a measurement, so it carries a method and a population.

Sources. arXiv (quant-ph, q-fin), Google Scholar, IEEE Xplore, Springer Link, and the reference lists of every paper already included. *Queries.* The cross product of QAOA, quantum approximate optimization, quantum alternating operator ansatz, quantum annealing with portfolio optimization, portfolio selection, portfolio rebalancing, each also run with “benchmark” and with “QUBO”. *Window.* Through 2026-08-10. *Inclusion,* all four required: the paper runs QAOA — including constraint-preserving mixers in the alternating-operator family — or quantum annealing; on a portfolio selection or rebalancing problem; on a quantum device or simulator; and reports at least one solution-quality metric of its own. *Exclusion.* Surveys, quantum-inspired or classical-only methods, other quantum algorithm families (VQE, quantum-walk, Grover-type search), and documents that propose a reporting protocol rather than use one, which are read with the reporting proposals instead. One paper is named but unread for want of institutional access.

Selection flow. The searches surfaced 26 candidate papers, of which twelve were excluded. Eleven fall under four disjoint grounds: four are surveys that report no runs of their own; three use a quantum algorithm outside the QAOA and annealing families (a VQE with a Dicke ansatz, a quantum-walk optimizer, a Grover-type search); one is quantum-inspired with no device or simulator run; and three propose a reporting protocol rather than use one, so they are read with the reporting proposals instead. The twelfth is named rather than silently dropped: a conference paper we could not obtain for want of institutional access. It is outside the population, so every count below is over the remaining fourteen, and would be over fifteen if that paper proved to qualify.

Coding. One coder, reading each paper’s formulation rather than its abstract, recording one row per paper with a verbatim quote for each cell that becomes a criticism. There is no second coder — this is a single-author study — so the per-paper cells and their quotes are committed, and a re-check of them found two errors, both in our own favour, which are recorded with their corrections. That is the accountability a second coder

would otherwise provide, offered as an artifact rather than as an assurance.

That protocol identifies fourteen benchmarks — [8]–[15], [22]–[27]. We read each paper’s *formulation* rather than its prose, recorded one row per paper, and committed the per-paper cells with quotes alongside the script that derives the counts below. What follows is therefore a claim about this population; whether it generalizes to the wider quantum-optimization literature is untested.

Counts derived from the committed census, not written out by hand.

The first row counts papers reporting *some* normalised quality number, not papers using this paper’s denominator: of the four, one normalises over the feasible set [9], one takes a relative distance to a brute-force optimum [22], one a regret against an offline reference [8] and one a min-max over the compared methods [15]. None divides by an energy carrying the penalty constant. These benchmarks have largely abandoned the gap, and the replacements are all different — a Wasserstein distance, a time-to-solution hit rate, a perturbation-tolerant success rate, a ground-state success probability, a feasibility rate, an arctan-compressed unnormalized energy, a realized Sharpe, a raw objective value. So **the third row is the one that matters**: proposals exist that would restore comparability, and none of the seven benchmarks that could have used them cites one. Not citing is what the row counts; whether a paper departs from a proposal’s content is a separate question that the census answers per cell. Sec. VI-A of the main text sets what the results add to what they ask for.

S-II. THE SINGLE-INSTANCE BENCHMARK

The benchmark the paper’s single-instance measurements come from ($N = 8$ – 30 , seed 42, $p = 2$), kept because it locates the simulator envelope (Sec. V-A) and because every later table qualifies it. After the conditioning fix of Sec. VI-C, $\langle H_{\text{norm}} \rangle$ converges within ~ 20 ADAM steps at every size (Fig. S1); classical runtime is flat in the seconds range while QAOA grows from 30 s at $N = 20$ to 3.14 h at $N = 30$ (Fig. S2), with no crossover anywhere in the simulable range. $N = 30$ at single precision uses 41.3 of 49 GB of VRAM (adjoint differentiation keeps ≈ 3 statevector copies); double precision would need ≈ 51.6 GB and does not fit; $N = 50$ needs $2^{50} \times 8 \text{ B} \approx 9 \text{ PB}$ and is infeasible for any statevector simulator. The $\langle H_{\text{norm}} \rangle$ column is the objective the optimizer actually descended: at $N = 20$ the run reached -0.91 against -5.00

TABLE S1
WHAT FOURTEEN PUBLISHED QAOA/ANNEALING PORTFOLIO BENCHMARKS REPORT.

Reported quantity	Papers reporting it
An optimality gap, approximation ratio or regret	4 of 14
Task counts, billed device time, or monetary cost	0 of 14
Citation of [19], [20] or [21]	0 of the 7 published after [19], [20] existed (0 of 14 overall)
Repeats per configuration	3 of 14
A finance/domain metric as a <i>result</i>	6 of 14
Solution degeneracy, observed	4 of 14

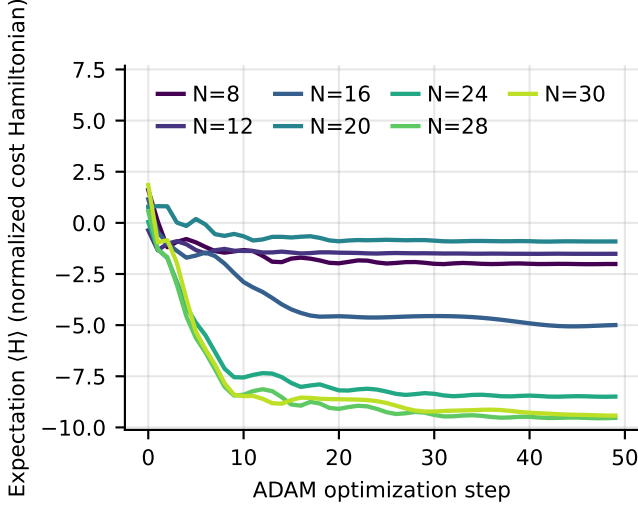


Fig. S1. $\langle H_{\text{norm}} \rangle$ vs. ADAM step, one curve per N (Table S2).

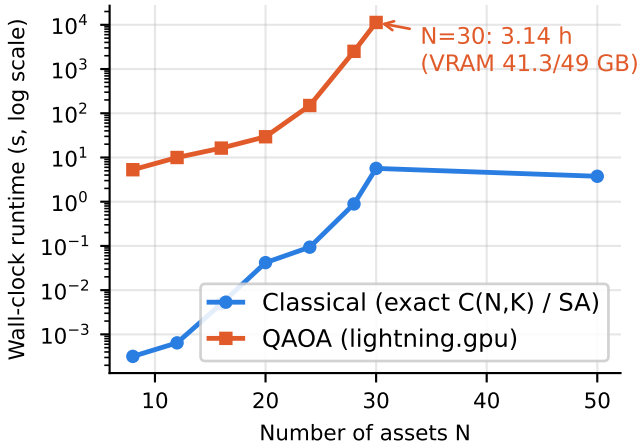


Fig. S2. runtime vs. N , classical (exact enumeration to $N = 30$, annealing at $N = 50$) vs. QAOA (log y).

one row above and -8.49 one row below and still reports a 0.053% gap, the second-best number in the column, because best-of-1,000 recovered a good sample from a poor state. Sec. V-B measures that effect across the cross. The reported gap per size is Fig. S3 and the Sharpe ratio of the selected portfolio Fig. S4.

Sharpe ratios agree exactly wherever the energies do ($N \leq 16$) and diverge far more than the gap suggests once QAOA is merely near-optimal: at $N = 20$ the 0.053% gap comes with

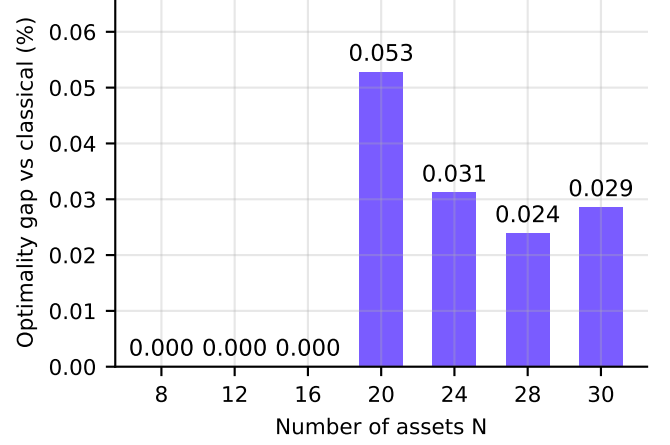


Fig. S3. reported gap vs. N (Table S2).

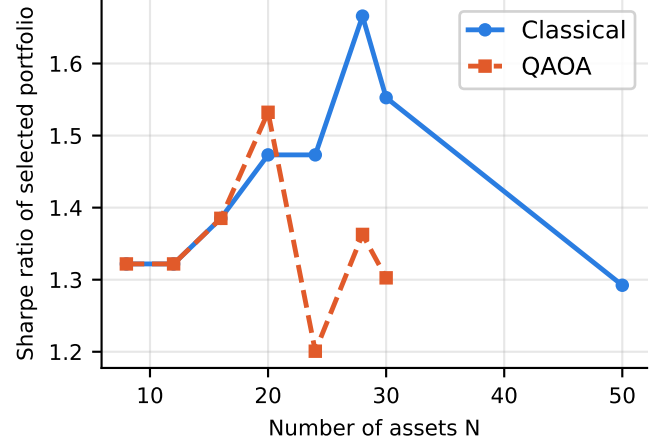


Fig. S4. Sharpe ratio of the selected portfolio vs. N (Table S2).

a *higher* Sharpe than the energy-optimal selection (1.53 vs. 1.47), while at $N = 24, 28$ and 30 gaps of 0.024 – 0.031% come with Sharpe ratios 15 – 22% below it (1.20 vs. 1.47 , 1.36 vs. 1.67 , 1.30 vs. 1.55 ; Fig. S4). Sec. S-III measures why.

S-III. THE METRIC ON THE REFERENCE INSTANCES

Sec. V-A reports the deflation, the band a 0.1% offset-normalized gap admits, and the all-states ratio of [19] on the reference instances. This section holds the tables behind those numbers: the census by exact enumeration (Table S3), the relocation of the single-seed QAOA solutions of Table S2

TABLE S2
QAOA VS. CLASSICAL ON THE REFERENCE INSTANCES (INSTANCE 0; SEED 42, P = 2, SINGLE PRECISION).

N	K	Classical E	QAOA E	Offset-normalized gap g_{off}	$\langle H_{\text{norm}} \rangle$	QAOA runtime	Feasible
8	2	-36.25	-36.25	0.000%	-2.01	5.3 s	✓
12	2	-52.25	-52.25	0.000%	-1.51	10 s	✓
16	3	-153.27	-153.27	0.000%	-5.00	17 s	✓
20	4	-336.15	-335.97	0.053%	-0.91	29 s	✓
24	4	-400.15	-400.03	0.031%	-8.49	150 s	✓
28	5	-725.17	-724.99	0.024%	-9.53	42 min	✓
30	6	-1115.96	-1115.64	0.029%	-9.42	3.14 h	✓
50	10	-5098.08	—	— (infeasible)	—	—	✓ (classical)

in the full ranking (Table S4), the threshold sweep (Table S5), the census under a universe-wide normalization (Table S6), the same QAOA solution under three denominators as the penalty weight is moved to its exact threshold (Table S7) and the all-states ratio (Table S8). Fig. S5 shows the Sharpe ratios spanned by the 0.1% band and the single-seed QAOA solutions under both denominators.

“Spread” is the range of the penalty-free objective over the feasible set; “deflation” is $|E^*|$ divided by it; the last two columns describe the solutions lying within a 0.1% reported gap of the optimum.

The deflation factor rises monotonically with N, as the growth of $A \cdot K^2$ against the measured spread predicts. Its consequence for the numbers in Table S2 is direct: **the QAOA gaps of 0.024–0.053% are 5.7–9.2% of the objective’s own spread over the feasible set.** At $N = 30$ the QAOA solution ranks 844th of 593,775 — the top 0.14%, which sounds excellent — and its Sharpe ratio is 16.1% below the optimum’s.

Why 0.1%, and what happens either side of it. The band is a step function of the threshold, so a single threshold invites the reading that it was chosen to flatter the argument. It was not; it is the conservative choice. Sweeping the threshold over the same enumeration:

Fraction of the feasible set inside the band, by size. The figure quoted in the text uses 0.1%.

At $N = 30$ the band holds 22 solutions at 0.01%, 146,484 at 0.1%, and 562,148 — 94.7% of everything feasible — at 0.2%. A benchmark quoting 0.2% rather than 0.1% would have a stronger version of this paper’s finding, not a weaker one. What the sweep shows is that there is no plateau to stand on: between 0.05% and 0.2% the band goes from 1.5% of the feasible set to 94.7%, so in exactly the range where reported gaps live, the metric has almost no resolution. The threshold also matters more as N grows — 3.6% of the feasible set at $N = 8$ against 24.7% at $N = 30$ for the same 0.1% — which is the same constant appearing in the band count rather than in the denominator.

Is the degeneracy an artefact of the normalization? Sec. III scales μ and Σ by *this instance’s* max-abs value, so a reader can reasonably ask whether the near-optimal band looks crowded because every instance has been squeezed into the same objective range. We repeated the enumeration with the scale held fixed instead — taken once from the fifty-ticker universe and applied to every subset, which lowers $|\hat{\mu}|$ and $|\hat{\Sigma}|$ below one and therefore lowers A as well.

Left: each instance scaled by its own max-abs value, as everywhere else in this paper. Right: one scale from the fifty-ticker universe, applied to every size.

It is not an artefact. Deflation survives at $109\text{--}262\times$ against $121\text{--}279\times$, eight to ten per cent lower and rising with N for the same reason. The band survives too, and at the two largest sizes it is *wider* under the fixed scale — 27.1% of the feasible set at $N = 30$ against 24.7%. The Sharpe range inside the band widens at every size, its floor falling from 0.172 to 0.070 at $N = 30$. If anything the per-instance normalization the rest of this paper uses understates the effect, so the choice is conservative rather than convenient.

The penalty weight sets the scale of the reported gap. The heuristic $A = 2 \cdot s_{\text{obj}} \cdot N + 1$ of Sec. III-B exceeds the exact threshold A_{crit} by two to three orders of magnitude on the reference instances; Table S7 scores the same QAOA solution at the weight used, at the threshold, and against the objective’s own spread. For $N > 20$ the threshold is the bounded A_{safe} , which understates the overshoot.

The last three columns are the same QAOA solution measured three ways. The heuristic penalty reports 0.024–0.053%; the threshold penalty reports 4.0–8.9% for those identical portfolios; and the penalty-independent measure against the objective’s own spread gives 5.7–9.2%. The second and third are different denominators — $|f^* - A_{\text{crit}} \cdot K^2|$ against the spread Δ_F — and they are close where $A_{\text{crit}} \cdot K^2$ happens to be near Δ_F ($N = 20, 24$) and apart where it is not (4.24% against 7.97% at $N = 30$, where the bounded A_{safe} also overstates the threshold). What the table shows is that the discrepancy between the first column and either of the others is the overshoot; it does not show the two alternative denominators agreeing, and they need not.

$AR = (C_{\text{max}} - C)/(C_{\text{max}} - C_{\text{min}})$, with C_{max} taken over all 2^N assignments. For $N \leq 20$, C_{max} is the exact maximum over the enumerated 2^N states (it falls at cardinality N); at $N = 30$ it is the energy of the all-ones assignment, which we have not proven maximal, so that row is a lower bound on C_{max} and the ratios there are, if anything, understated. “Worst feasible” is the highest-objective member of the enumerated feasible set — the worst answer a solver could return while still satisfying the constraint.

The metric does not degrade as the penalty grows; **at the penalty our formulation already uses, it has already collapsed.** At $N = 16$ the QAOA solution and the worst feasible portfolio in the entire set score 1.000000 and 0.999560 — the whole feasible set occupies four decimal places, and the

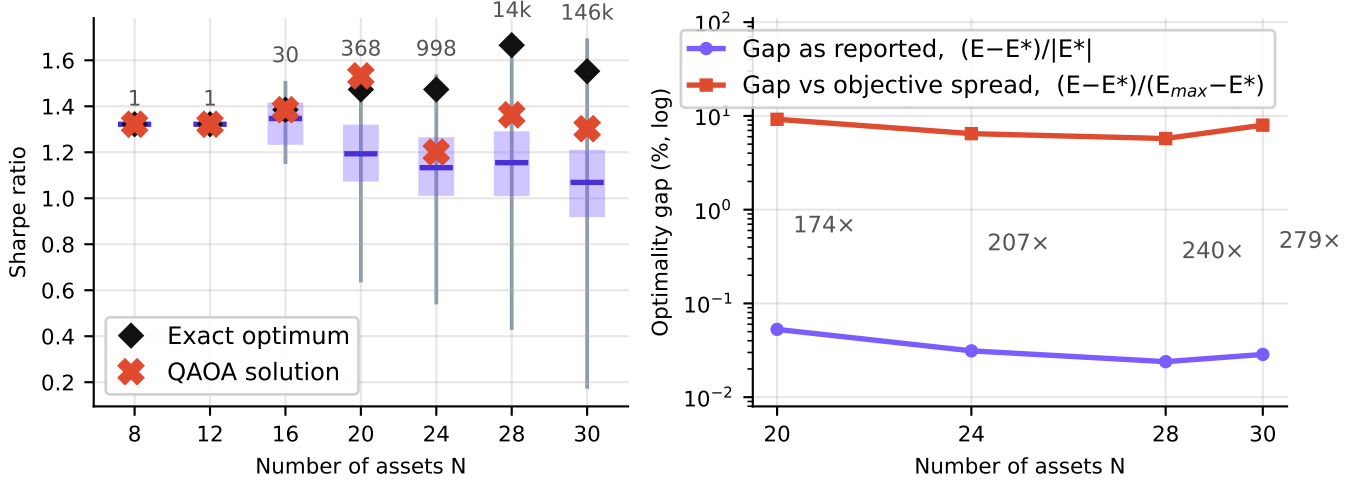


Fig. S5. Sharpe ratios spanned by every feasible solution within a 0.1% offset-normalized gap, per N , and the single-seed QAOA solutions under both denominators.

TABLE S3
WHAT THE REPORTED GAP DENOMINATES (EXACT ENUMERATION, SEED 42).

N	$C(N,K)$	Spread	$ E^* $	Deflation	In 0.1% band	Sharpe there
16	560	1.271	153.3	121×	30 (5.4%)	1.149 – 1.511
20	4,845	1.930	336.2	174×	368 (7.6%)	0.634 – 1.539
24	10,626	1.930	400.2	207×	998 (9.4%)	0.539 – 1.539
28	98,280	3.023	725.2	240×	13,737 (14.0%)	0.428 – 1.707
30	593,775	4.000	1116.0	279×	146,484 (24.7%)	0.172 – 1.695

TABLE S4
THE SINGLE-SEED QAOA SOLUTIONS OF TABLE S2 RELOCATED IN THE FULL RANKING.

N	Gap as reported	Gap vs objective spread	Rank	Percentile	Sharpe (optimum → QAOA)
20	0.0528%	9.20%	80 / 4,845	1.65%	1.473 → 1.532 (+4.0%)
24	0.0312%	6.47%	43 / 10,626	0.40%	1.473 → 1.201 (−18.5%)
28	0.0239%	5.74%	111 / 98,280	0.11%	1.666 → 1.363 (−18.2%)
30	0.0286%	7.97%	844 / 593,775	0.14%	1.553 → 1.303 (−16.1%)

TABLE S5
THE NEAR-OPTIMAL BAND AS A FUNCTION OF THE REPORTED-GAP THRESHOLD.

Threshold	$N = 8$	$N = 12$	$N = 16$	$N = 20$	$N = 24$	$N = 28$	$N = 30$
0.01%	3.6%	1.5%	0.5%	0.1%	0.1%	0.0%	0.0%
0.05%	3.6%	1.5%	2.0%	1.5%	1.2%	1.1%	1.5%
0.1%	3.6%	1.5%	5.4%	7.6%	9.4%	14.0%	24.7%
0.2%	3.6%	4.5%	18.0%	42.5%	57.1%	80.8%	94.7%
0.5%	21.4%	25.8%	80.7%	99.8%	100%	100%	100%
1.0%	50.0%	81.8%	100%	100%	100%	100%	100%

TABLE S6
THE CENSUS UNDER TWO NORMALIZATIONS.

N	Deflation, per-instance	Deflation, global	Band, per-instance	Band, global	Sharpe range, global
16	121×	109×	5.4%	3.4%	0.716 – 1.511
20	174×	162×	7.6%	7.1%	0.274 – 1.539
24	207×	193×	9.4%	9.2%	0.274 – 1.539
28	240×	222×	14.0%	15.5%	0.148 – 1.707
30	279×	262×	24.7%	27.1%	0.070 – 1.695

solutions Sec. V-A shows spanning Sharpe ratios from 0.172 to 1.695 are, on this axis, indistinguishable. Multiplying A by ten shrinks the remaining separation by a further order of

magnitude. This is not a criticism of [19], whose calibration problems have no feasibility constraint to violate; it is a statement that the one prescribed comparable metric is practically

TABLE S7
THE PENALTY WEIGHT SILENTLY RESCALES THE REPORTED GAP.

N	K	A used	A_{crit} (A_{safe} for $N > 20$)	Overshoot	Gap at A used	Gap at A_{crit}	Gap vs objective spread
8	2	9	0.0107	840×	0.0000%	0.00%	0.00%
12	2	13	0.0107	1213×	0.0000%	0.00%	0.00%
16	3	17	0.0441	385×	0.0000%	0.00%	0.00%
20	4	21	0.1148	183×	0.0528%	8.93%	9.20%
24	4	25	0.1148	218×	0.0312%	6.28%	6.47%
28	5	29	0.1657	175×	0.0239%	4.02%	5.74%
30	6	31	0.2099	148×	0.0286%	4.24%	7.97%

TABLE S8
THE ALL-STATES APPROXIMATION RATIO OF [19] EVALUATED ON OUR OWN INSTANCES.

N	Penalty A	AR, our QAOA solution	AR, worst feasible
16	17 (heuristic)	1.000000	0.999560
16	170 (10×	1.000000	0.999956
20	21 (heuristic)	0.999967	0.999642
20	210 (10×	0.999997	0.999964
30	31 (heuristic)	0.999982	0.999777
30	310 (10×	0.999998	0.999978

non-discriminative for the constrained family studied here.

S-IV. THIRTY INSTANCES PER SIZE, AND CLASSICAL ANNEALING

Sec. V-A summarises the instance axis and the two annealing arms. Table S9 is the census repeated over thirty instances per size; Table S10 the annealing-on-the-feasible-set arm beside the seed-42 QAOA arm; Table S11 annealing on the penalty-encoded QUBO; Table S12 the seed-42 QAOA arm in full, with the gap/Sharpe association of Fig. S7. Fig. S6 shows the deflation and the Sharpe spread inside the 0.1% band over the thirty instances, with the reference instance marked.

Medians over the thirty unless noted; “Ref. inst.” is the deflation of the reference instance (instance 0, the one the rest of the paper uses), “in 0.1% band” counts feasible solutions within a 0.1% reported gap, and “overshoot” is the heuristic penalty divided by the minimum that keeps the optimum feasible.

Both mechanisms survive as distributions rather than anecdotes. The median deflation rises monotonically with N , as the growth of $A \cdot K^2$ against the spread predicts, and no instance at any size escapes it — the *smallest* deflation we observed anywhere is $36\times$ at $N = 12$. The median penalty overshoot stays between $103\times$ and $179\times$ at every size, so it is a property of the heuristic rather than of a particular covariance matrix. The reference instance is the exception, and for a reason worth stating: its A_{crit} is 0.0107 at $N = 8$ and $N = 12$, small enough that the same heuristic weight overshoots it $840\times$ and $1213\times$ (Table S7). Overshoot is a ratio against a threshold that can itself be near zero, so at small N it is far more dispersed than the deflation factor, which is why the two are quoted differently — $121\text{--}279\times$ for deflation on the reference instance against $36\text{--}411\times$ across instances, but $148\text{--}1213\times$ for overshoot on the reference instance against a $103\text{--}179\times$ median. At $N = 12$, 9 of 30 instances have $A_{\text{crit}} = 0$ — their unconstrained optimum already has cardinality K — and for

them the overshoot is undefined rather than large, so the $N = 12$ median is taken over the other 21. (The exact all-states A_{crit} of Sec. III-B is used here and everywhere in Sec. V; the bounded A_{safe} would count seven, because it is 0.008–0.009 on two instances whose exact A_{crit} is 0.)

The reference instance is unremarkable in deflation — near the median at every size — but it is *favourable* in the quantity that matters to a reader. Ranked by the Sharpe spread inside a 0.1% gap it is below the median at *every* size (18th, 17th, 37th, 13th, 3rd and 3rd percentile for $N = 12 \dots 30$), and at $N = 28$ and $N = 30$ it has the narrowest spread of all thirty instances. Fig. S6 shows both quantities across the sweep: the deflation factor, which no instance escapes and whose smallest observed value anywhere is $36\times$, and the Sharpe spread inside the 0.1% band, where the reference instance sits at the bottom of the range rather than in the middle of it. The median instance at $N = 30$ has 261,202 feasible solutions inside that band, 44% of the entire feasible set, spanning a Sharpe range of 142% of the optimum’s. The single-instance results understate the effect rather than exaggerate it.

Simulated annealing on the feasible set. The gap’s denominator, the band and the objective/Sharpe decoupling are properties of the QUBO and its scoring rather than of the solver. Simulated annealing with cardinality-preserving swaps, five seeds on each of the same 30 instances at each size (900 runs), scored against the same enumerated optima, reached the exact optimum in 895 of 900 runs; all five misses are at $N = 30$, the largest gap is 0.0022% and the worst Sharpe deviation 3.4%. The two arms are not on matched budgets (50 ADAM steps against 50,000 annealing moves). That arm searches the feasible set directly and never enters the penalty landscape, so it bounds what a competent classical baseline achieves on the *task*, not what a stochastic solver scores on the *encoding*; Table S11 runs the same annealer on the penalty-encoded Q with single-bit flips.

Same Q as the QAOA arm, same enumerated optimum; moves are single-bit flips, so the search walks the penalty landscape rather than the feasible set.

Every one of the fifteen runs returned a feasible portfolio, and every one of them reproduces the pattern the QAOA arm shows: a reported gap under one per cent alongside a Sharpe ratio tens of per cent below the optimum’s, and a rank deep inside the feasible set. At $N = 20$, seed 46 reports 0.33% while sitting 4,307th of 4,845. So the four effects that do not depend on the sampler — the denominator, the near-degeneracy, the objective/Sharpe decoupling and the instance axis — are properties of the encoding, and a classical solver

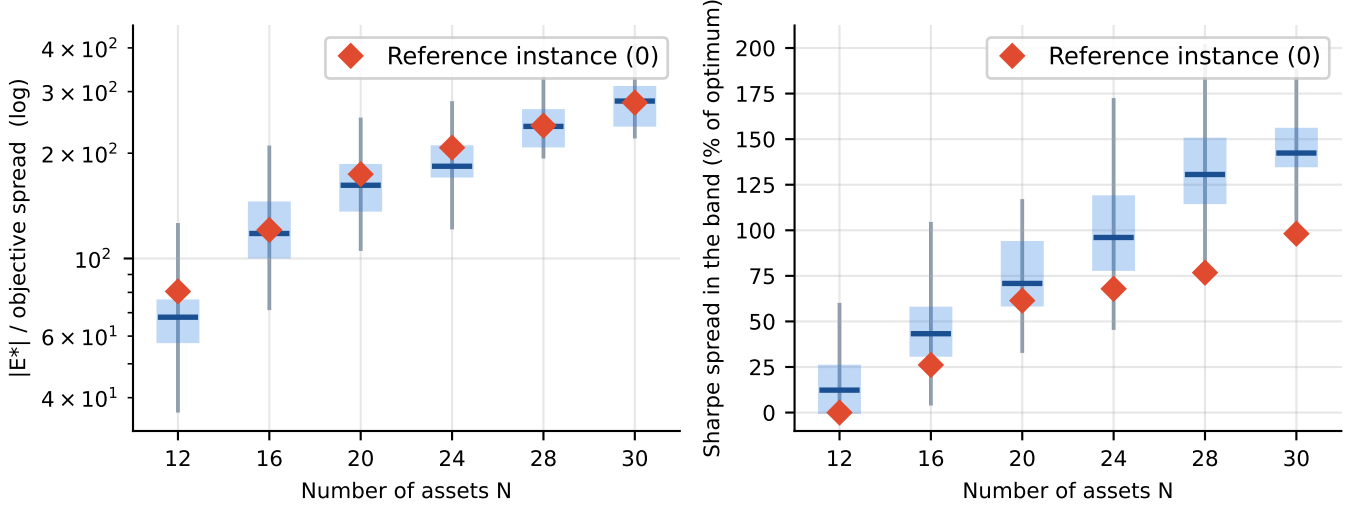


Fig. S6. deflation and the Sharpe spread inside the 0.1% band over 30 instances per size, with the reference instance marked.

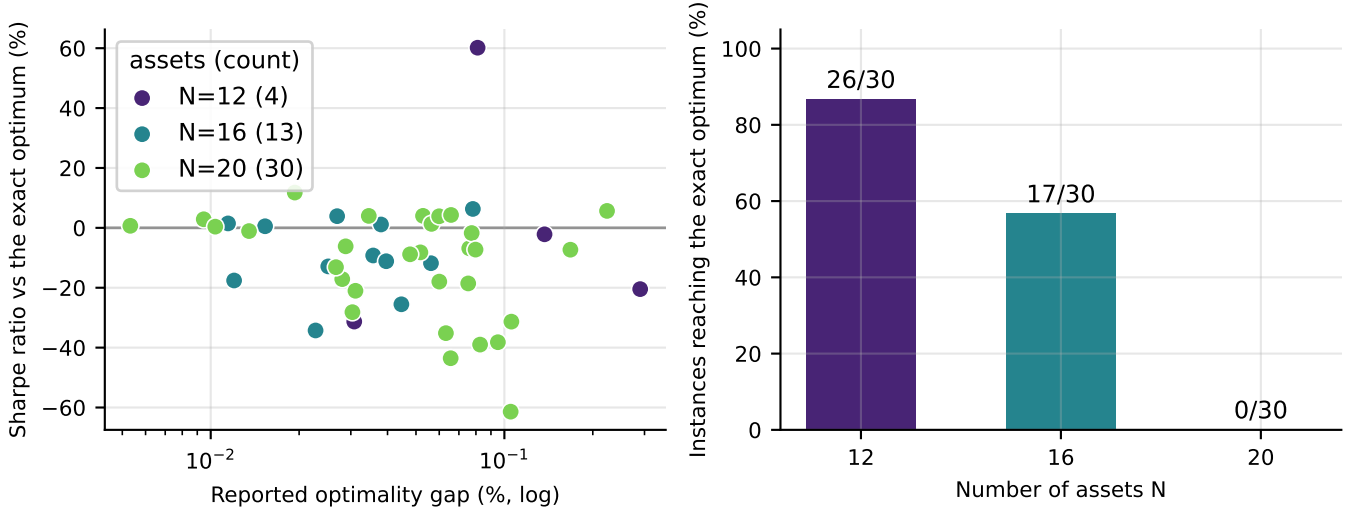


Fig. S7. every sub-optimal QAOA solution of the seed-42 instance sweep, reported gap against Sharpe deviation; and the exact-optimum rate per size.

TABLE S9
THE CENSUS REPEATED OVER 30 INSTANCES PER SIZE.

N	Deflation med. (range)	Ref. inst.	In 0.1% band	Sharpe spread	Overshoot
12	68× (36–126)	81×	2	12%	179×
16	118× (71–210)	121×	22	43%	136×
20	162× (105–253)	174×	354	71%	118×
24	184× (121–282)	207×	1,052	96%	150×
28	239× (193–332)	240×	26,670	131%	124×
30	282× (220–411)	279×	261,202	142%	103×

TABLE S10
THE SAME 30 INSTANCES PER SIZE, SCORED AGAINST THE SAME ENUMERATED OPTIMA.

N	Simulated annealing	median runtime	QAOA (p = 2, 50 steps)	median runtime
12	150/150	3.4 s	26/30	9.9 s
16	150/150	3.4 s	17/30	16.4 s
20	150/150	3.4 s	0/30	29.0 s
24	150/150	3.4 s	—	—
28	150/150	3.4 s	—	—
30	145/150	3.4 s	—	—

TABLE S11
CLASSICAL ANNEALING ON THE PENALTY-ENCODED QUBO, FIVE SEEDS PER SIZE.

N	Exact optimum	Reported gap	Δ Sharpe	Worst rank
12	1 of 5	0.00 – 0.59%	0 to −64.2%	24 of 66
16	0 of 5	0.14 – 0.52%	−3.7 to −36.3%	466 of 560
20	0 of 5	0.10 – 0.33%	−23.5 to −49.9%	4,307 of 4,845

scored the same way inherits all of them. The pattern belongs to the encoding scored this way, not to the sampler.

Running QAOA over the same instances qualifies Table S2’s headline. We repeat the QAOA configuration of Table S2 — $p = 2$, 50 ADAM steps, 1000 shots, initial-angle seed 42 — on 30 instances at each of $N = 12, 16, 20$, and locate every returned solution in the exhaustively ranked feasible set.

“Gap/spread” restates the reported gap against the objective’s own range over the feasible set; “rank” is the solution’s position in that set; Δ Sharpe is relative to the exact optimum’s.

Every one of the 90 runs returned a feasible lowest-energy shot; Sec. V-B shows that this is a property of seed 42 rather than of the configuration — across the five seeds of the cross, 14 of 150 $N = 16$ runs at A_{heur} returned no feasible bitstring at all. Table S2’s “exact agreement for $N \leq 16$ ” is true of the reference instance and holds in 26 of 30 instances at $N = 12$ but only 17 of 30 at $N = 16$; at $N = 20$ it never happens. Reporting it from one instance overstated it.

The 47 runs that missed carry the paper’s sharpest evidence that an objective gap is not a portfolio guarantee, and the direction of the error is not fixed. At $N = 12$ every miss is *rank 2* — the second-best portfolio in the entire feasible set — and the four of them sit at Δ Sharpe = **+60.2%**, **−2.2%**, **−20.5%** and **−31.3%**: the same rank, in the same instance family, is worth anything from a 60% improvement to a third of the Sharpe ratio. At $N = 16$ the worst miss reports a gap of 0.0228%, while its Sharpe falls from 1.089 to 0.716, a 34.3% loss, at rank 4 of 560. At $N = 20$ one instance reports 0.1050% and loses 61.4%. Across all 47 misses, 17 have a Sharpe ratio *above* the exact optimum’s, so the correct statement is not that small gaps hide losses but that they carry no sign.

The association has to be characterised carefully, and it is weaker than “uncorrelated” but not absent. All correlations below are against $|\Delta$ Sharpe|, the magnitude of the deviation, not its signed value. Pooled over the 47 misses, Pearson $r = 0.17$ ($p = 0.25$) but Spearman $\rho = 0.34$ ($p = 0.019$). Part of that is a size confound — $N = 20$ supplies both the larger gaps (median 0.058%) and the larger Sharpe deviations (median 7.3%) — but not all of it: *within* $N = 20$ alone, Spearman $\rho = 0.43$ ($p = 0.019$, $n = 30$), against Pearson $+0.22$ ($p = 0.24$) on the same 30 points. At the two smaller sizes the rank association is absent or negative ($\rho = -0.60$ at $N = 12$, $n = 4$; $+0.06$ at $N = 16$, $n = 13$; neither significant). The two p -values being equal to three decimals is a coincidence of rounding, not a transcription error: they are 0.0187 and 0.0189 under SciPy’s asymptotic t approximation. Reporting only the Pearson coefficients within size would have dismissed a relationship that a reader recomputing from the committed CSV will find. Two things keep it from rescuing the metric: it is one of three within-size tests, so its $p = 0.019$ becomes

0.057 under a Bonferroni correction, and a rank correlation of 0.43 on 30 points is a loose ordering, not a predictor. The rank of the returned solution within the feasible set does no better, at Spearman 0.13 against $|\Delta$ Sharpe| ($p = 0.40$).

The defensible claim is therefore narrower than “the gap says nothing”, and still damaging: **the reported gap predicts neither the sign of the Sharpe deviation nor, usefully, its size.** The sign result is unconditional — 17 of the 47 misses beat the exact optimum’s Sharpe ratio, so a small gap does not even tell a reader which direction they moved. The magnitude survives only as a weak rank tendency at one size, next to individual pairs that contradict it outright: 0.0228% against a 34.3% loss, and four solutions of identical rank spanning +60.2% to −31.3%. A quantity that ranks 30 points loosely while mispredicting individual ones by that much is not a portfolio guarantee, which is what Fig. S7 shows directly.

S-V. THE CROSS IN DETAIL (E1, E2, E5)

The full tables behind Sec. V-B: the decomposition of the paired gap ratio at every size (Table S13), the four samplers at every shot budget (Table S14), the two-window comparison in full (Table S15), the disagreement of the two success rules over both thresholds (Table S16), the split of P_τ into feasible mass and selection within it (Table S17), the conditional distance of every run from uniform on F (Table S18), and the design subsets on which the main contrast depends (Table S19).

Pairs with a zero gap on either side, or without a feasible shot, are counted and excluded from the ratio. The identity holds pair by pair, and a median does not carry it: the median of a product is not the product of the medians. The medians are therefore reported as levels and the decomposition is taken in the geometric mean, which is multiplicative over the same pairs because log turns the product into a sum.

At $N = 20$ the A_{margin} run reports a gap that is 29 times larger in the median, and 27.8 times larger in the geometric mean, than the A_{heur} run on the same instance and seed. The decomposition assigns 36.6 of that to the denominator and 0.76 to the solution, which is *better*: it is better in the median too (0.81) and on 59% of pairs. The reported gap moves because the denominator moves, with the solution pulling the other way.

The two uniform samplers are instance properties. The last three columns count runs whose optimised state beats the uniform-over- F sampler, decided on the one-shot probability so the count is the same at every S : at $\tau = 1\%$, at the exact optimum and at $\tau = 5\%$.

Instances 0–9 $\times$ seeds 42–46 $\times$ A_{heur} , A_{margin} , 300 runs per window; “2023–26” is the paper’s window (2023-08-06 – 2026-08-05) and “2020–23” the adjacent earlier one (2020-08-05 – 2023-08-05). Levels are medians over the 50 runs per

TABLE S12
QAOA OVER 30 INSTANCES PER SIZE (P = 2, SEED 42).

N	Exact optimum	Gap med. (max)	Gap/spread med.	Rank med. (max)	Δ Sharpe med. (worst)
12	26/30	0.0000% (0.2900%)	0.00%	1 (2)	+0.0% (−31.3%)
16	17/30	0.0000% (0.0781%)	0.00%	1 (6)	+0.0% (−34.3%)
20	0/30	0.0582% (0.2236%)	8.94%	52 (504)	−7.3% (−61.4%)

TABLE S13
DECOMPOSITION OF THE $A_{\text{margin}} / A_{\text{heur}}$ RATIO OF REPORTED GAPS.

N	Pairs used (zero gap / no feasible shot)	Medians: g_{off} ratio / solution (IQR) / denominator	Geometric mean: g_{off} ratio = solution $\times$ denominator	Solution part < 1
12	16 (134 / 0)	29.4 / 0.54 (0.42–0.91) / 46.8	34.7 = 0.71 $\times$ 49.1	75%
16	49 (80 / 21)	58.1 / 1.00 (0.49–2.18) / 48.2	45.8 = 1.00 $\times$ 45.6	43%
20	128 (21 / 1)	29.0 / 0.81 (0.39–1.34) / 40.0	27.8 = 0.76 $\times$ 36.6	59%

TABLE S14
 $Q_\tau(S)$ FOR FOUR SAMPLERS AT A_{heur} (MEDIAN OVER INSTANCES OF THE WITHIN-INSTANCE MEDIAN OVER SEEDS; $\tau = 1\%$ OF Δ_F).

N	S	Optimised QAOA	Unoptimised (same initial angles)	Uniform over F	Uniform over 2^N	QAOA beats uniform-F, $\tau = 1\%$	exact optimum	$\tau = 5\%$
12	10	0.026	0.017	0.142	0.0024	0 / 150	0 / 150	0 / 150
12	1,000	0.928	0.817	1.000	0.217	0 / 150	0 / 150	0 / 150
16	100	0.040	0.006	0.164	0.0015	0 / 150	0 / 150	0 / 150
16	1,000	0.336	0.062	0.833	0.015	0 / 150	0 / 150	0 / 150
16	10,000	0.983	0.470	1.000	0.142	0 / 150	0 / 150	0 / 150
20	100	0.006	0.002	0.040	0.0002	0 / 150	0 / 150	0 / 150
20	1,000	0.059	0.019	0.338	0.0019	0 / 150	0 / 150	0 / 150
20	10,000	0.458	0.171	0.984	0.019	0 / 150	0 / 150	0 / 150

cell; paired rows are $A_{\text{margin}} - A_{\text{heur}}$ reduced to an instance median and bootstrapped over the ten instances (95% interval), with the count of instances on which A_{margin} is better (ties in parentheses); g_{off} and g_F are of the best feasible shot. The ratio rows give two aggregations of the $A_{\text{margin}} / A_{\text{heur}}$ g_{off} ratio: the medians of the ratio, of its solution part and of its denominator part, and the geometric-mean decomposition, which is multiplicative over the same pairs (Sec. V-B). The E2 row counts A_{heur} runs whose optimised state beats the uniform-over-F sampler at $S = 1,000$.

Sec. V-B reports that 76% of $N = 20$ heuristic-weight runs pass a 0.1% offset-normalized gap and miss a 1% feasible-range criterion. Both rules have a threshold, and one pair of thresholds is one point on a surface. Table S16 reads both rules at four gap thresholds and three quality thresholds on the same runs.

The disagreement is monotone in the gap threshold and concentrated on the heuristic weight, which is what the deflation account predicts: at A_{heur} and $N = 20$ it runs 0.06, 0.49, 0.76, 0.89 across gap thresholds of 0.01%, 0.05%, 0.1% and 0.2%, while at A_{margin} it stays between 0.03 and 0.07 at every one of them. The 0.1% figure is neither the largest available nor a threshold at which something special happens; the tightest threshold nearly removes the disagreement, because a gap of 0.01% under a 174-fold deflation is still a demanding requirement.

Share of runs in which exactly one of the two rules passes. Columns are the offset-normalized gap threshold; “quality” is the feasible-range criterion $g_F \leq \tau$, with the exact optimum as its limiting case.

Feasible mass and selection within it. The one-shot quality

probability factors as $P_\tau = P_F \cdot \Pr[g_F \leq \tau \mid x \in F]$, and the second factor can be compared with what an indifferent draw from the feasible set would give, $|F_\tau|/|F|$. Their ratio R_τ is 1 for a state that finds the feasible set but sorts nothing inside it and greater than 1 for one that concentrates on the good part. It is computed per run and then summarised; dividing the medians of P_τ and P_F would be a different quantity.

Table S17 measures it, and the heuristic weight comes out at $R_\tau = 1.00$: 0.998, 0.998 and 1.009 at $N = 12, 16$ and 20 , with interquartile ranges of 0.02 or less. Conditioned on landing in the feasible set, the optimised state at A_{heur} picks a solution the way a uniform draw from F would. The margin weight does sort, at the two smaller sizes — $R_\tau = 1.40$ (IQR 1.04–1.97) at $N = 12$ and 1.55 (1.21–2.04) at $N = 16$, above 1 on 114 and 125 of 150 runs — and stops doing so at $N = 20$ (1.01, IQR 0.80–2.10).

R_τ checks the mass on one subset of F only. A ratio of 1 is consistent with the conditional distribution being uniform on F and also with its being anything else that puts the same mass in F_τ , so the uniformity Sec. V-B reports is established by the conditional distance of Table S18, not by this table.

Medians over the 150 runs of each cell. $|F_\tau|/|F|$ is the share of the feasible set within $\tau = 1\%$ of the feasible range, a property of the instance. R_τ is the per-run ratio of the conditional rate to that share.

A. The conditional distribution on the feasible set

For every E1 run the artifact holds the exact probability $p(x)$ of every feasible bitstring under the final state and under the initial state. The distance of the conditional distribution from uniform on F ,

TABLE S15
THE E1 BLOCK ON TWO PRICE WINDOWS (E5).

Quantity	N = 12, 2023–26	N = 12, 2020–23	N = 16, 2023–26	N = 16, 2020–23	N = 20, 2023–26	N = 20, 2020–23
Δ_F , median	0.705	0.942	1.296	1.736	2.195	2.710
$A_{\text{heur}} / A_{\text{crit}}$, median (instances with $A_{\text{crit}} = 0$)	92 (4)	199 (3)	139 (0)	115 (0)	99 (0)	92 (0)
$A_{\text{heur}}: Q_\tau(1000)$	0.926	0.933	0.336	0.395	0.060	0.098
$A_{\text{heur}}: g_{\text{off}} (\%) / g_F (\%)$	0.000 / 0.00	0.000 / 0.00	0.023 / 2.19	0.017 / 1.64	0.043 / 6.48	0.036 / 4.64
$A_{\text{heur}}: \text{pass } g_{\text{off}} \leq 0.1\%, \text{ fail } g_F \leq 1\%$	0.04	0.04	0.47	0.49	0.74	0.66
$A_{\text{heur}}: \text{runs with no feasible shot}$	0	0	3	7	0	0
Paired $Q_\tau(1000)$ [CI]; better of 10	+0.020 [+0.000, +0.067]; 8	+0.010 [+0.000, +0.061]; 9	+0.041 [+0.004, +0.062]; 10	+0.057 [+0.029, +0.074]; 10	+0.090 [+0.067, +0.202]; 10	+0.122 [+0.062, +0.219]; 10
Paired g_{off} (pt) [CI]; better of 10	0 [0, 0]; 0 (9)	0 [0, 0]; 0 (10)	+0.28 [−0.00, +1.41]; 2 (2)	+0.19 [−0.01, +0.76]; 4 (0)	+0.63 [+0.41, +0.95]; 1 (0)	+0.52 [+0.20, +0.90]; 0 (0)
Paired g_F (pt) [CI]; better of 10	0 [0, 0]; 0 (9)	0 [0, 0]; 0 (10)	0.00 [−2.31, +0.07]; 4 (4)	−0.22 [−2.05, +0.13]; 5 (3)	−1.75 [−4.49, −0.73]; 9 (0)	−1.93 [−3.80, 0.00]; 7 (2)
g_{off} ratio / solution / denominator, medians (pairs)	27.5 / 0.50 / 54.2 (6)	44.4 / 0.92 / 48.4 (3)	60.8 / 1.26 / 47.8 (20)	44.7 / 1.62 / 34.3 (17)	26.3 / 0.74 / 39.5 (42)	18.0 / 0.67 / 27.8 (44)
g_{off} ratio = solution $\times$ denominator, geometric means	25.5 = 0.50 $\times$ 51.3	18.7 = 0.38 $\times$ 49.6	42.7 = 0.98 $\times$ 43.7	69.1 = 2.05 $\times$ 33.7	22.3 = 0.64 $\times$ 34.6	12.4 = 0.45 $\times$ 27.4
E2: A_{heur} beats uniform-F at S = 1,000	0 / 50	0 / 50	0 / 50	0 / 50	0 / 50	0 / 50

TABLE S16
DISAGREEMENT BETWEEN THE TWO SUCCESS RULES, OVER BOTH THRESHOLDS.

N	A	quality rule	0.01%	0.05%	0.1%	0.2%
12	A_{heur}	1%	0.01	0.01	0.03	0.06
12	A_{heur}	5%	0.03	0.01	0.01	0.04
12	A_{heur}	exact	0.01	0.03	0.05	0.07
12	A_{margin}	1%	0.01	0.01	0.01	0.01
12	A_{margin}	5%	0.05	0.05	0.05	0.04
12	A_{margin}	exact	0.00	0.00	0.00	0.01
16	A_{heur}	1%	0.05	0.33	0.46	0.51
16	A_{heur}	5%	0.24	0.04	0.16	0.22
16	A_{heur}	exact	0.07	0.35	0.48	0.54
16	A_{margin}	1%	0.04	0.04	0.03	0.01
16	A_{margin}	5%	0.26	0.26	0.24	0.23
16	A_{margin}	exact	0.00	0.00	0.01	0.03
20	A_{heur}	1%	0.06	0.49	0.76	0.89
20	A_{heur}	5%	0.32	0.11	0.38	0.52
20	A_{heur}	exact	0.09	0.52	0.79	0.93
20	A_{margin}	1%	0.07	0.07	0.06	0.03
20	A_{margin}	5%	0.43	0.43	0.43	0.39
20	A_{margin}	exact	0.00	0.00	0.01	0.04

TABLE S17
 P_τ SPLIT INTO FEASIBLE MASS AND SELECTION WITHIN THE FEASIBLE SET.

N	A	P_F	$\Pr[g_F \leq \tau \mid x \in F]$	$ F_\tau / F $	R_τ (IQR)	$R_\tau > 1$
12	A_{heur}	0.179	$1.52 \cdot 10^{-2}$	$1.52 \cdot 10^{-2}$	1.00 (0.99–1.01)	68/150
12	A_{margin}	0.328	$2.20 \cdot 10^{-2}$	$1.52 \cdot 10^{-2}$	1.40 (1.04–1.97)	114/150
16	A_{heur}	0.229	$1.78 \cdot 10^{-3}$	$1.79 \cdot 10^{-3}$	1.00 (1.00–1.00)	50/150
16	A_{margin}	0.301	$3.01 \cdot 10^{-3}$	$1.79 \cdot 10^{-3}$	1.55 (1.21–2.04)	125/150
20	A_{heur}	0.167	$4.07 \cdot 10^{-4}$	$4.13 \cdot 10^{-4}$	1.01 (0.99–1.01)	90/150
20	A_{margin}	0.303	$4.19 \cdot 10^{-4}$	$4.13 \cdot 10^{-4}$	1.01 (0.80–2.10)	85/150

$$D_{\text{cond}} = \frac{1}{2} \sum_{x \in F} \left| \frac{p(x)}{P_F} - \frac{1}{|F|} \right|,$$

is 0 for a uniform conditional distribution and $1 - 1/|F|$ for a point mass; it is undefined at $P_F = 0$, and runs with P_F below 10^{-9} are counted rather than normalised (none occurred). Two reference distances are printed beside it: the same distance for the *initial* state of the run, and the mean distance a 1,000-shot empirical distribution drawn from uniform-on-F would show at that $|F|$ (200 Monte Carlo draws), which is what uniform looks like at the experiment’s own shot resolution — 0.10 at

$|F| = 66$ and 0.81 at $|F| = 4,845$, so the question could not have been settled from the shots and is settled from the exact probabilities. Table S18 gives the distribution over runs per cell; the per-run file is `paper01_e1_conditional_tv.csv`.

At A_{heur} the optimised state is close to uniform on F — within 0.04 at $N = 12$ and 0.01 at $N \geq 16$ in every run, 0.001–0.002 in the median. The initial state is already close in the median but not in every run (as far as 0.41 at $N = 20$ for some seeds), so the optimisation removes the ordering the random angles carried rather than adding one (Table S34 pairs each run’s initial and final state). Every other weight departs further from uniform: the closer A is to A_{crit} the more, and A_{margin}

sits between. D_{cond} carries no direction; whether the departure favours the good solutions is read from R_r in Table S17 (above 1 at A_{margin} for $N \leq 16$).

Median, IQR and range over the runs of each cell; “initial” is the median for the unoptimised state at the same angles; “uniform, 1,000 shots” is what a finite-shot draw from the uniform distribution would show; “point mass” is the distance of a single-bitstring state. Runs with $P_F = 0$ would be counted in the “undefined” column; there were none.

B. Which parts of the design the main contrast depends on

The interval Sec. V-B quotes for $A_{margin} - A_{heur}$ is over the thirty instances of a size, with the same five initial-angle seeds inside every one and instance 0 — the reference instance — deliberately among them. Table S19 re-runs the same paired contrast on subsets of that design. The subsets are of the design and not of the outcome: nothing is selected on the value being summarised.

Three of them change nothing. Dropping instance 0 moves the $N = 20$ median from +0.086 to +0.083 and the $N = 16$ median from +0.0397 to +0.0390. Splitting the $N = 12$ instances by whether A_{crit} is zero gives +0.011 on the twenty-one positive ones and +0.013 on the nine zero ones, both with intervals clear of zero. Dropping any one seed leaves the $N = 20$ median between +0.044 and +0.112, improved on 30 of 30 instances every time.

The fourth changes the sign. Conditioned on a *single* seed, one of the five reverses the contrast at each of the two larger sizes: seed 45 at $N = 16$ (−0.157, better on 0 of 30 instances) and seed 44 at $N = 20$ (−0.096, 0 of 30). Which seed it is differs by size, and at $N = 12$ none reverses though two have intervals containing zero. This is the paper’s own finding about initial angles arriving in its statistics: the instance-level interval is an interval for the contrast *averaged over these five initial angles*, and it is not an interval for the contrast at an angle drawn from any wider distribution. Readers wanting the latter need a design that samples initialisations, which this one does not.

Instance-level median difference with its 95% bootstrap interval, and the count of instances on which A_{margin} is better. “ $A_{crit} = 0$ ” exists only at $N = 12$.

S-VI. DEPTH, OPTIMIZER BUDGET AND THE OPTIMIZER CONTROLS (E3)

The single-seed depth sweep (Table S20) and the steps $\times$ p $\times$ seed grid (Table S21, Fig. S10) that Sec. V-C summarises; the optimizer controls in full (Table S22, Fig. S11), the dispersion behind each signed median (Table S23), the stopping-criterion diagnostic at both precisions, and the multistart re-analysis at equal shots (Table S24).

Depth, optimizer budget, and what the reported gap actually measures (single-seed grid).

Runtime is essentially linear in p (Fig. S9, right), as expected: depth multiplies the per-step circuit work while the 50-step optimizer budget stays fixed. Quality, however, is not monotone (Fig. S9, left). Going from $p = 1$ to $p = 2$ removes the gap outright at $N = 12$ and $N = 16$ — both reach the proven

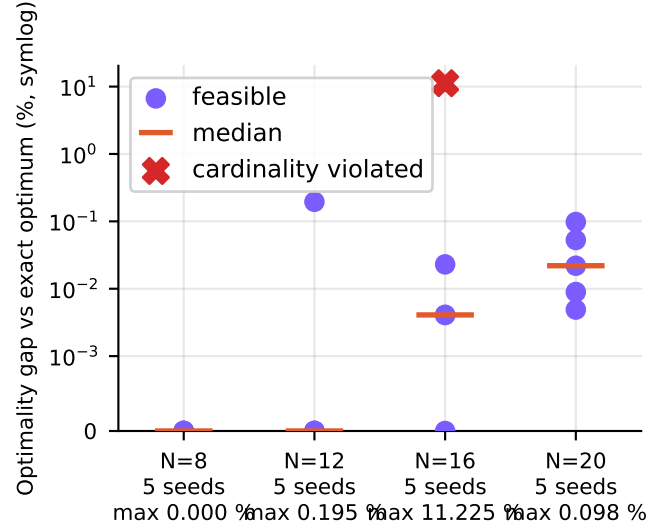


Fig. S8. the five-seed sweep (a sub-block of E1): gap across initial-angle seeds (symlog), the infeasible run marked separately.

optimum — and cuts it $5.0\times$ at $N = 20$, which justifies the $p = 2$ used throughout Sec. V-A. Beyond $p = 2$ the ordering is not stable and does not repeat across sizes: $p = 3$ improves $N = 20$ (0.0528% $\rightarrow$ 0.0089%) but loses the exact hit at $N = 16$ (0.0000% $\rightarrow$ 0.0516%), while $p = 4$ recovers part of that loss at $N = 16$ (0.0427%) and regresses $N = 20$ to 0.0563%. Depth beyond $p = 2$ changes which basin the run lands in without systematically improving it, at roughly double the runtime. Table S20 holds the ADAM budget at 50 steps, so it cannot by itself say whether the $p \geq 3$ non-monotonicity is the ansatz or simply too few optimizer iterations for the larger parameter vector. We resolve that with a grid.

The steps $\times$ p $\times$ seed grid. We ran QAOA over $N \in 12, 16, 20 \times p \in 1, 2, 3, 4 \times$ steps $\in 50, 100, 400 \times$ five seeds (180 runs), recording both the reported gap and the optimized objective $\langle H_{norm} \rangle$ (Fig. S10). Two things separate cleanly.

The optimizer is not budget-starved. The optimized objective saturates by 50–100 ADAM steps and does not improve with an $8\times$ larger budget; several cells are marginally *worse* at 400 steps because ADAM with a fixed 0.1 step size wanders near the minimum rather than settling. Table S21 shows seed 42.

So the $p \geq 3$ non-monotonicity in Table S20 is not a shortage of iterations *for this optimizer setting*: ADAM at a fixed step of 0.1 does not recover it with eight times the budget. Whether a smaller step or a quasi-Newton method would be what E3 below tests at $p = 2$.

What does move is the initial angle. At a fixed, saturated budget the depth that lands in a bad basin changes with N : at $N = 16$ it is $p = 3$ ($\langle H_{norm} \rangle = -2.89$, against $p = 2$ ’s -5.07), while at $N = 20$ it is $p = 2$ (−0.91, against $p = 3$ ’s -3.87). A monotone property of the ansatz would not flip which depth is unlucky from one size to the next; a $2p$ -dimensional non-convex landscape with a fixed random start would. This is the seed variance of Sec. V-B appearing on the depth axis — more parameters give the optimizer more bad basins to fall into —

TABLE S18
DISTANCE OF THE CONDITIONAL DISTRIBUTION ON F FROM UNIFORM, PER CELL.

N	A	Runs (undefined)	P_F	D_{cond} median (IQR)	D_{cond} min – max	Initial-state D_{cond}	Uniform, 1,000 shots	Point mass
12	A = 0	45 (0)	0.199	0.554 (0.480–0.603)	0.303 – 0.690	0.456	0.102	0.985
12	$1 \cdot A_{crit}$	105 (0)	0.257	0.382 (0.208–0.498)	0.053 – 0.747	0.302	0.102	0.985
12	$2 \cdot A_{crit}$	105 (0)	0.277	0.308 (0.148–0.456)	0.036 – 0.969	0.238	0.102	0.985
12	$5 \cdot A_{crit}$	105 (0)	0.244	0.207 (0.070–0.389)	0.012 – 0.891	0.172	0.102	0.985
12	$10 \cdot A_{crit}$	105 (0)	0.190	0.108 (0.030–0.267)	0.008 – 0.844	0.090	0.102	0.985
12	A_{margin}	150 (0)	0.328	0.173 (0.089–0.393)	0.027 – 0.701	0.195	0.102	0.985
12	A_{heur}	150 (0)	0.179	0.002 (0.001–0.013)	0.000 – 0.038	0.003	0.102	0.985
16	$1 \cdot A_{crit}$	150 (0)	0.245	0.369 (0.201–0.545)	0.072 – 0.844	0.354	0.301	0.998
16	$2 \cdot A_{crit}$	150 (0)	0.275	0.192 (0.117–0.321)	0.035 – 0.888	0.369	0.301	0.998
16	$5 \cdot A_{crit}$	150 (0)	0.359	0.106 (0.050–0.209)	0.020 – 0.749	0.342	0.301	0.998
16	$10 \cdot A_{crit}$	150 (0)	0.309	0.080 (0.022–0.136)	0.003 – 0.465	0.122	0.301	0.998
16	A_{margin}	150 (0)	0.301	0.139 (0.087–0.214)	0.014 – 0.464	0.377	0.301	0.998
16	A_{heur}	150 (0)	0.229	0.001 (0.001–0.003)	0.000 – 0.009	0.002	0.301	0.998
20	$1 \cdot A_{crit}$	150 (0)	0.238	0.494 (0.239–0.614)	0.051 – 0.867	0.354	0.813	1.000
20	$2 \cdot A_{crit}$	150 (0)	0.224	0.198 (0.080–0.401)	0.008 – 0.827	0.228	0.813	1.000
20	$5 \cdot A_{crit}$	150 (0)	0.233	0.061 (0.022–0.114)	0.005 – 0.479	0.095	0.813	1.000
20	$10 \cdot A_{crit}$	150 (0)	0.219	0.036 (0.014–0.067)	0.003 – 0.366	0.054	0.813	1.000
20	A_{margin}	150 (0)	0.303	0.090 (0.022–0.148)	0.006 – 0.387	0.145	0.813	1.000
20	A_{heur}	150 (0)	0.167	0.002 (0.001–0.003)	0.001 – 0.005	0.004	0.813	1.000

TABLE S19
THE $A_{margin} - A_{heur}$ CONTRAST IN $Q_\tau(1000)$ UNDER SUBSETS OF THE DESIGN.

Subset	N = 12	N = 16	N = 20
all	+0.011 [+0.005, +0.034]; 27/30	+0.040 [+0.031, +0.061]; 30/30	+0.086 [+0.071, +0.116]; 30/30
without instance 0	+0.011 [+0.007, +0.026]; 26/29	+0.039 [+0.029, +0.061]; 29/29	+0.083 [+0.067, +0.111]; 29/29
instance 0 only	+0.074; 1/1	+0.043; 1/1	+0.231; 1/1
$A_{crit} > 0$ instances	+0.011 [+0.000, +0.026]; 18/21	+0.040 [+0.030, +0.061]; 30/30	+0.086 [+0.072, +0.116]; 30/30
$A_{crit} = 0$ instances	+0.013 [+0.001, +0.063]; 9/9	—	—
seed 42 only	+0.067 [−0.053, +0.073]; 20/30	+0.037 [+0.025, +0.043]; 26/30	+0.088 [+0.080, +0.153]; 30/30
seed 43 only	+0.095 [+0.074, +0.173]; 30/30	+0.003 [+0.002, +0.004]; 30/30	+0.001 [+0.001, +0.002]; 30/30
seed 44 only	+0.000 [+0.000, +0.000]; 26/30	+0.384 [+0.359, +0.414]; 30/30	−0.096 [−0.119, −0.065]; 0/30
seed 45 only	+0.006 [−0.130, +0.074]; 17/30	−0.157 [−0.162, −0.152]; 0/30	+0.130 [+0.096, +0.155]; 30/30
seed 46 only	+0.011 [+0.010, +0.012]; 29/30	+0.367 [+0.302, +0.428]; 30/30	+0.124 [+0.109, +0.170]; 30/30
without seed 42	+0.006 [+0.006, +0.039]; 29/30	+0.188 [+0.159, +0.208]; 30/30	+0.055 [+0.047, +0.070]; 30/30
without seed 43	+0.006 [+0.003, +0.022]; 24/30	+0.194 [+0.154, +0.222]; 30/30	+0.112 [+0.083, +0.130]; 30/30
without seed 44	+0.040 [+0.010, +0.057]; 24/30	+0.022 [+0.017, +0.029]; 27/30	+0.112 [+0.083, +0.130]; 30/30
without seed 45	+0.031 [+0.006, +0.041]; 29/30	+0.217 [+0.172, +0.228]; 30/30	+0.045 [+0.039, +0.065]; 30/30
without seed 46	+0.035 [+0.008, +0.056]; 24/30	+0.022 [+0.017, +0.029]; 27/30	+0.044 [+0.040, +0.061]; 30/30

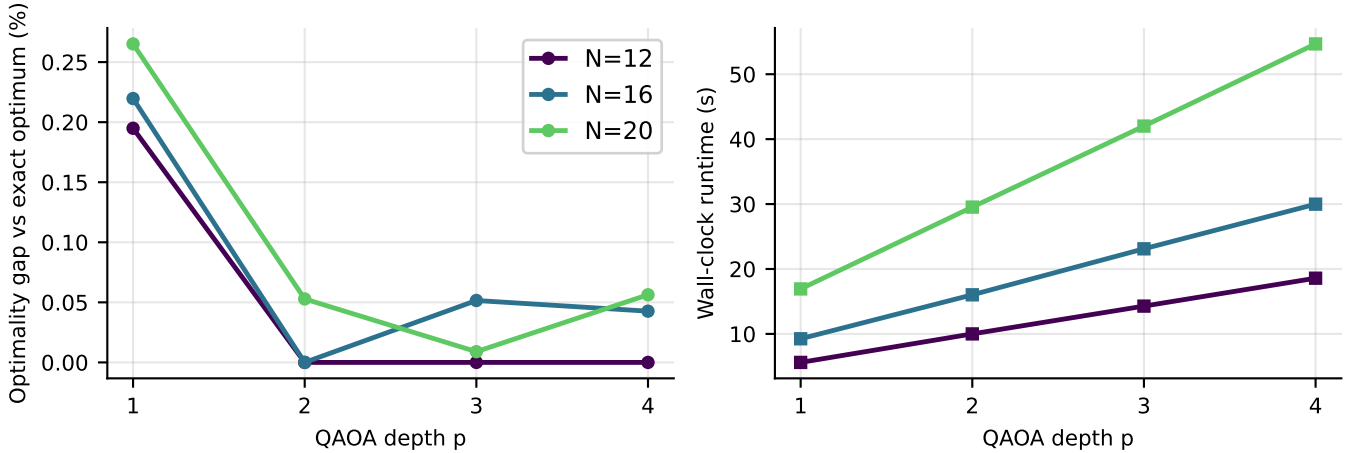


Fig. S9. gap and runtime vs. QAOA depth p (Table S20).

which is consistent with an initialisation effect; the runs do not separate initialisation, local minima and expressivity, and the cause is not identified. A fixed-step ADAM that stalls in the same basin at 50 and at 400 steps has not been shown to explore, so on the grid alone “not budget” meant “not iteration count”, not “not optimizer”; E3 closes that gap for $p = 2$.

The reported gap can hide all of this. Because the gap is the best of 1000 final samples, it is a minimum over samples, not the quantity the optimizer descended. A run can barely optimize and still report a small gap: at $N = 20$, steps = 400, $p = 2$ the optimizer effectively failed ($\langle H_{norm} \rangle = -0.91$, scarcely below $p = 1$ ’s -0.63) yet its reported gap is 0.038%,

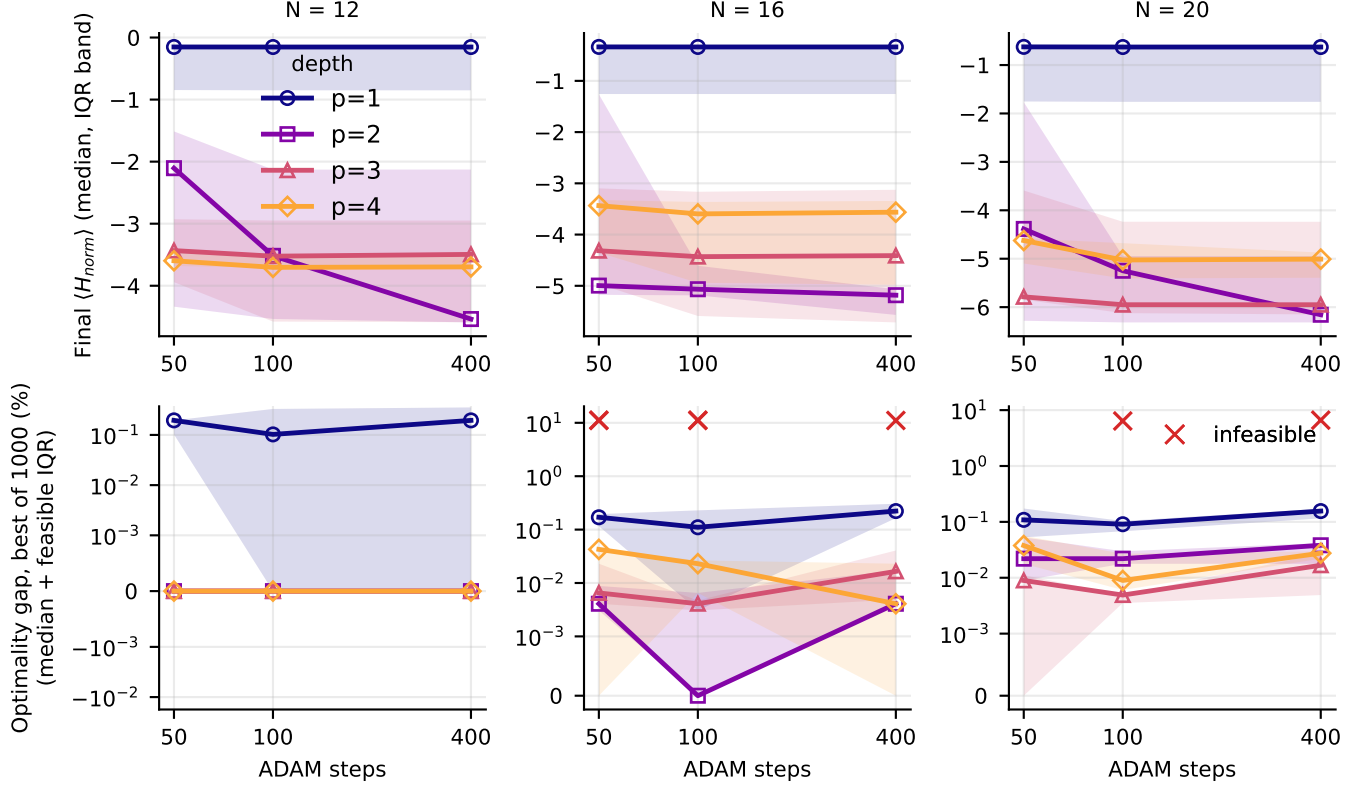


Fig. S10. the steps $\times$ p grid over $N \in 12, 16, 20$: optimised $\langle H_{\text{norm}} \rangle$ (top) and best-of-1,000 gap (bottom), median with IQR over five seeds.

TABLE S20
OPTIMALITY GAP AND RUNTIME VS. QAOA DEPTH p (SEED 42, EXACT CLASSICAL REFERENCE).

N	$p = 1$	$p = 2$	$p = 3$	$p = 4$	runtime $p = 1 \rightarrow 4$
12	0.1949%	0.0000%	0.0000%	0.0000%	5.6 s $\rightarrow$ 18.6 s
16	0.2197%	0.0000%	0.0516%	0.0427%	9.3 s $\rightarrow$ 30.0 s
20	0.2651%	0.0528%	0.0089%	0.0563%	16.9 s $\rightarrow$ 54.7 s

TABLE S21
OPTIMIZED EXPECTATION AT EACH OPTIMIZER BUDGET, SEED 42, THREE CELLS OF THE GRID.

final $\langle H_{\text{norm}} \rangle$	steps = 50	100	400
$N = 16, p = 2$	-4.9959	-5.0669	-5.0674
$N = 16, p = 4$	-4.3513	-4.9416	-4.9796
$N = 20, p = 3$	-3.5908	-3.9047	-3.8730

indistinguishable from the well-optimized $p = 4$ in the same cell ($\langle H_{\text{norm}} \rangle = -5.01$, gap 0.022%). Fig. S10 therefore plots both rows — optimized $\langle H_{\text{norm}} \rangle$ on top, reported gap below — precisely so the disagreement is visible. The methodological point is that **a QAOA benchmark that reports only a best-of-shots gap can present a failed optimization as a success**; the optimized objective must be reported alongside it. At this optimizer setting $p = 2$ is where depth stops helping monotonically; beyond it the iteration count is saturated, and the remaining variation is not attributed to a cause.

A. The optimizer controls in full

Levels are medians over the 50 runs per cell; the paired columns are the instance-level median difference from the baseline with its 95% bootstrap interval, and the count of instances (of 10) on which the control reached a lower $\langle H_{\text{norm}} \rangle$. Budget is the median number of gradient evaluations actually used; every ADAM step is one objective and one gradient evaluation, and for L-BFGS-B the objective and gradient counts were equal in every one of the 300 runs, $n_f = n_{\nabla}$, line-search evaluations included (each trial point evaluates both), so the two budgets are comparable as printed. The last column is the median optimisation wall-clock. “Optimum hit” counts runs whose best feasible shot is the exact optimum over *all fifty* runs of the cell, with the number that produced no feasible shot in parentheses; a run that returns nothing usable is a failure to hit, not an absent observation. Conditioning on a feasible shot instead would raise $N = 16$, A_{heur} , L-BFGS-B from 0.34 to 0.40 — eight of its fifty runs found no feasible shot, against three of the baseline’s — and widen a margin of three runs over the baseline (0.28) into one of ten points (0.30 conditioned the

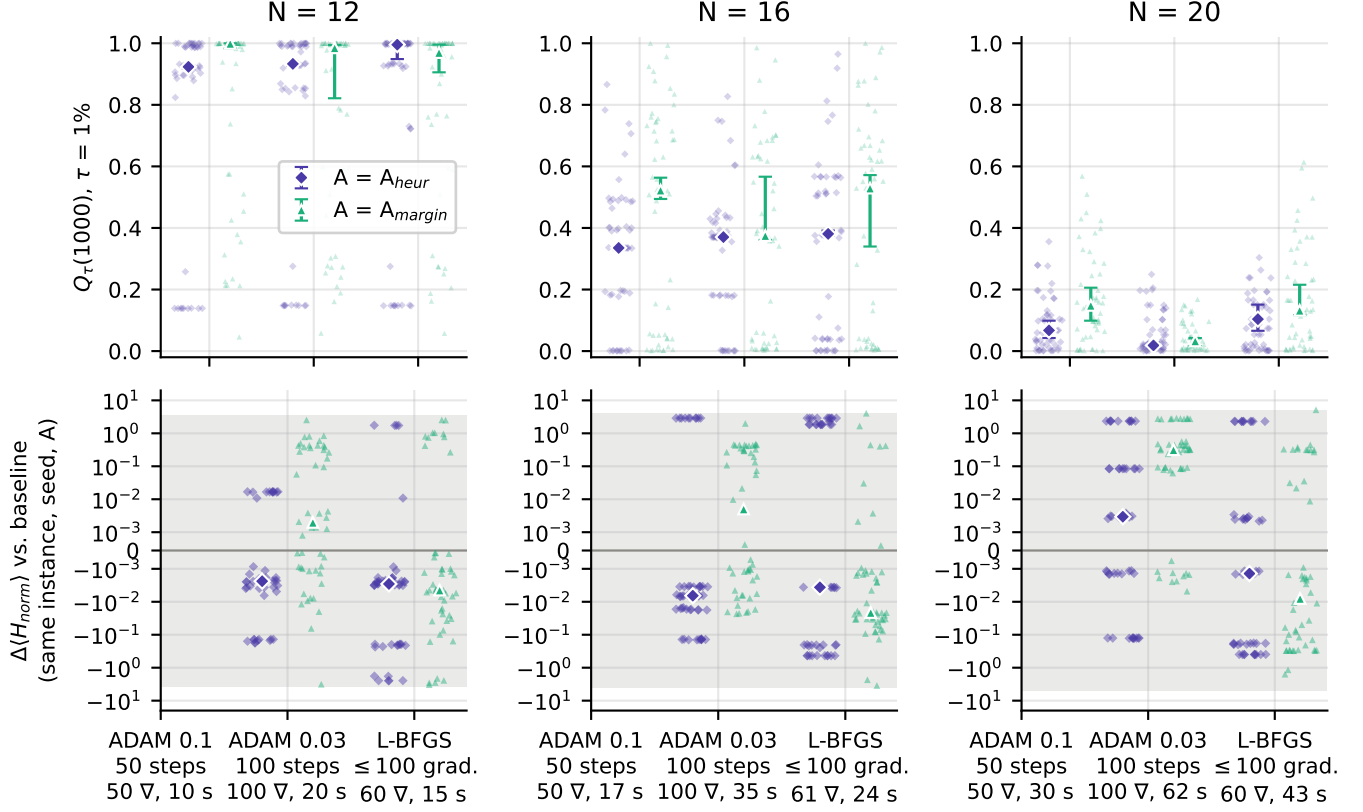


Fig. S11. E3. Top: $Q_\tau(1000)$ under each optimizer setting for A_{heur} and A_{margin} , every run as a point. Bottom: the paired change in $\langle H_{\text{norm}} \rangle$ against the baseline (symlog), with the baseline’s within-instance seed spread as a grey band and the budget actually used under each tick.

same way). “Predicted”, in the same layout (hits, then non-feasible runs in parentheses), is what the cell’s fifty recorded exact states give for the same two counts under independent draws, $\Sigma_j [1 - (1 - P_{0,j})^{1000}]$ hits and $\Sigma_j (1 - P_{F,j})^{1000}$ runs without a feasible shot, with P_0 the one-shot probability of the exact optimum (expected_optimum_hits and expected_no_feasible_shot_runs in paper01_e3_summary.csv); each run has its own probability, so the count is a sum of Bernoulli variables with different rates, not a binomial. The ADAM cells observe within four of the predicted hits and within three of the predicted no-feasible runs. This comparison is a check on the batch, not on the angles — it is what exposed the L-BFGS-B handoff defect of Sec. VI-C (the pre-fix arm observed 6 hits against 14.8 predicted at $N = 16$, A_{heur} , and 9 runs without a feasible shot against 0.2 at $N = 20$, A_{heur}) — and the argument check is the regression test named in Sec. S-VI. The re-run’s L-BFGS-B cells observe within 3.0 of the predicted hits and within 2.7 of the predicted no-feasible runs, as the ADAM cells do. ΔP_τ and $\Delta Q_\tau(1000)$ are separate measurements on the same pairs and are not interchangeable: Q is a batch quantity, $\partial Q_\tau(S)/\partial P_\tau = S(1 - P_\tau)(S-1)$, so a difference in P_τ too small to print can be worth a tenth of a batch. Table S23 adds the median $|\Delta|$, its 95th percentile and the spread behind each signed median. The $\langle H_{\text{norm}} \rangle$ levels at $N = 16$ and $N = 20$ sit on the boundary between two basins the runs fall into (≈ -2.2 and ≈ -5.1 at $N = 16$), so a shift

of 10^{-3} in individual runs can move the median across it; the paired column is the measurement.

Fig. S11 puts every paired change in $\langle H_{\text{norm}} \rangle$ next to the baseline’s own spread over the five seeds within an instance (the grey band is that seed range, not a factor contribution); the median of max – min over seeds is 3.4–5.5 at every N and both weights.

B. The spread behind each signed median

A signed median can be small because every change is small or because the changes cancel. Table S23 separates the two for the two acquisition quantities of Table S22: the typical size of a change, its 95th percentile, and how the ten instances split between better and worse. Read with the main table, it is what supports the claim that L-BFGS-B leaves P_τ alone (median $|\Delta P_\tau| \leq 3.8 \cdot 10^{-4}$ everywhere) while not leaving $Q_\tau(1000)$ alone (95th percentile of $|\Delta Q|$ reaches 0.079 at $N = 16$, A_{margin}), and that the one control with a large effect is ADAM at 0.03 at $N = 20$, A_{margin} .

Instance-level paired differences against the ADAM 0.1 $\times$ 50 baseline, ten instances per cell; the signed medians and their intervals are in Table S22; these are the columns that median cannot recover. “Better/worse/tied” counts instances, not runs.

TABLE S22
OPTIMIZER CONTROLS ON THE E3 BLOCK.

N	A	Setting	Budget (∇ , s)	$\langle H_{\text{norm}} \rangle$	$\Delta \langle H_{\text{norm}} \rangle$ vs. baseline lower in [CI];	P_τ	ΔP_τ [CI]	Q_τ (1000)	ΔQ_τ (1000) [CI]	Optimum hit / 50 (no feas.)	Predicted
12	A_{heur}	ADAM 0.1 $\times$ 50	50, 10	-2.121	—	$2.6 \cdot 10^{-3}$	—	0.926	—	38/50 (0)	39.4 (0.0)
12	A_{heur}	ADAM 0.03 $\times$ 100	100, 20	-2.124	$-2.4 \cdot 10^{-3}$ [-2.5, $-2.3 \cdot 10^{-3}$; 10/10	$2.7 \cdot 10^{-3}$	$-2.4 \cdot 10^{-6}$ [-9.4 $\cdot 10^{-5}$, $+1.1 \cdot 10^{-5}$]	0.934	+0.0010 [-0.0001, +0.0036]	42/50 (0)	39.2 (0.0)
12	A_{heur}	L-BFGS-B	52, 13	-3.937	$-3.0 \cdot 10^{-3}$ [-3.1, $-2.4 \cdot 10^{-3}$; 10/10	$5.4 \cdot 10^{-3}$	$+3.8 \cdot 10^{-4}$ [+1.2, $+3.9 \cdot 10^{-4}$	0.995	+0.0087 [+0.0044, +0.0094]	40/50 (0)	39.6 (0.0)
16	A_{heur}	ADAM 0.1 $\times$ 50	50, 17	-5.065	—	$4.1 \cdot 10^{-4}$	—	0.336	—	14/50 (3)	14.1 (5.8)
16	A_{heur}	ADAM 0.03 $\times$ 100	100, 35	-2.173	$-6.5 \cdot 10^{-3}$ [-6.8, $-6.2 \cdot 10^{-3}$; 10/10	$4.6 \cdot 10^{-4}$	$+2 \cdot 10^{-8}$ [-6 $\cdot 10^{-9}$, $+3 \cdot 10^{-8}$]	0.370	+0.0000 [-0.0000, +0.0000]	14/50 (4)	13.6 (5.8)
16	A_{heur}	L-BFGS-B	53, 21	-2.173	$-3.6 \cdot 10^{-3}$ [-3.7, $-3.6 \cdot 10^{-3}$; 9/10	$4.7 \cdot 10^{-4}$	$+1.6 \cdot 10^{-7}$ [+1.3, $+2.5 \cdot 10^{-7}$	0.377	+0.0002 [+0.0001, +0.0002]	17/50 (8)	14.8 (5.3)
20	A_{heur}	ADAM 0.1 $\times$ 50	50, 30	-4.556	—	$6.2 \cdot 10^{-5}$	—	0.060	—	2/50 (0)	2.1 (0.1)
20	A_{heur}	ADAM 0.03 $\times$ 100	100, 61	-2.211	+2.9 $\cdot 10^{-3}$ [+2.8, $+3.2 \cdot 10^{-3}$; 0/10	$2.5 \cdot 10^{-5}$	$-6.1 \cdot 10^{-6}$ [-9.3, $-2.7 \cdot 10^{-6}$	0.025	-0.0060 [-0.0092, -0.0027]	1/50 (0)	1.4 (0.2)
20	A_{heur}	L-BFGS-B	51, 36	-2.258	$-1.4 \cdot 10^{-3}$ [-1.5, $-1.2 \cdot 10^{-3}$; 10/10	$7.6 \cdot 10^{-5}$	$-1.5 \cdot 10^{-7}$ [-2.5, $-0.7 \cdot 10^{-7}$	0.074	-0.0002 [-0.0002, -0.0001]	1/50 (0)	2.2 (0.2)
12	A_{margin}	ADAM 0.1 $\times$ 50	50, 10	-4.069	—	$6.5 \cdot 10^{-3}$	—	0.998	—	38/50 (0)	39.5 (0.0)
12	A_{margin}	ADAM 0.03 $\times$ 100	100, 20	-3.796	$+1.3 \cdot 10^{-3}$ [-1 $\cdot 10^{-4}$, $+9.6 \cdot 10^{-2}$]; 4/10	$4.9 \cdot 10^{-3}$	$-3 \cdot 10^{-6}$ [-2.8, $+2.6 \cdot 10^{-4}$	0.993	-0.0000 [-0.0030, +0.0014]	36/50 (0)	37.7 (0.0)
12	A_{margin}	L-BFGS-B	65, 17	-4.114	$-4.3 \cdot 10^{-3}$ [-15.9, $-1.5 \cdot 10^{-3}$; 10/10	$5.4 \cdot 10^{-3}$	$+8.5 \cdot 10^{-5}$ [-2.5, $+4.7 \cdot 10^{-4}$	0.996	+0.0000 [-0.0029, +0.0010]	36/50 (0)	39.0 (0.0)
16	A_{margin}	ADAM 0.1 $\times$ 50	50, 17	-5.242	—	$7.4 \cdot 10^{-4}$	—	0.523	—	20/50 (2)	21.2 (4.0)
16	A_{margin}	ADAM 0.03 $\times$ 100	100, 35	-5.017	$-1.0 \cdot 10^{-3}$ [-1.2 $\cdot 10^{-3}$, $+5.5 \cdot 10^{-2}$]; 7/10	$4.6 \cdot 10^{-4}$	$-6.1 \cdot 10^{-5}$ [-10.4, $-1.9 \cdot 10^{-5}$	0.368	-0.0281 [-0.0710, -0.0128]	12/50 (3)	15.9 (1.6)
16	A_{margin}	L-BFGS-B	68, 27	-5.229	$-2.1 \cdot 10^{-2}$ [-2.4, $-1.2 \cdot 10^{-2}$; 10/10	$7.8 \cdot 10^{-4}$	$-3.0 \cdot 10^{-5}$ [-11.0, $+0.2 \cdot 10^{-5}$	0.540	-0.0160 [-0.0271, +0.0020]	17/50 (2)	19.1 (1.8)
20	A_{margin}	ADAM 0.1 $\times$ 50	50, 30	-6.139	—	$1.4 \cdot 10^{-4}$	—	0.130	—	6/50 (0)	4.3 (0.0)
20	A_{margin}	ADAM 0.03 $\times$ 100	100, 62	-5.826	+0.318 [+0.292, +0.331]; 0/10	$3.4 \cdot 10^{-5}$	$-1.2 \cdot 10^{-4}$ [-2.1, $-0.6 \cdot 10^{-4}$	0.033	-0.1060 [-0.1866, -0.0604]	0/50 (0)	1.1 (0.0)
20	A_{margin}	L-BFGS-B	63, 45	-5.900	$-4.1 \cdot 10^{-3}$ [-60.2, $-2.0 \cdot 10^{-3}$; 10/10	$1.4 \cdot 10^{-4}$	$+5 \cdot 10^{-7}$ [-1.0 $\cdot 10^{-5}$, $+5.5 \cdot 10^{-6}$]	0.127	+0.0005 [-0.0064, +0.0049]	3/50 (0)	4.3 (0.0)

C. What “stopped” means at single precision

The E3 block ran at single precision, as every run in this paper does, and recorded only how many gradient evaluations each run used — a count from which convergence cannot be inferred. SciPy’s default tolerances are $\text{ftol} = 2.2 \cdot 10^{-16}$ · $\text{factr} = 2.2 \cdot 10^{-9}$ on the relative reduction of f and $\text{gtol} = 10^{-5}$ on the

projected gradient; the first sits two orders of magnitude below the resolution of a single-precision expectation. Re-running a fixed block of the same configuration at both precisions (2 instances $\times$ 2 seeds $\times$ 2 weights at $N = 12$ and $N = 16$, 32 runs, local CPU) shows what the counts were hiding: in single precision **2 of 16 runs failed their line search and**

TABLE S23
DISPERSION OF THE E3 PAIRED DIFFERENCES.

N	A	Setting	median $ \Delta P_\tau $	p95 $ \Delta P_\tau $	P_τ better/worse	median $ \Delta Q $	p95 $ \Delta Q $	Q_τ better/worse
12	A_{heur}	ADAM 0.03×100	$1.3 \cdot 10^{-5}$	$2.7 \cdot 10^{-4}$	5/5	0.0012	0.0048	5/5
12	A_{heur}	L-BFGS-B	$3.8 \cdot 10^{-4}$	$7.4 \cdot 10^{-4}$	10/0	0.0087	0.0109	10/0
16	A_{heur}	ADAM 0.03×100	$2.0 \cdot 10^{-8}$	$4.3 \cdot 10^{-8}$	7/3	0.0000	0.0000	7/3
16	A_{heur}	L-BFGS-B	$1.7 \cdot 10^{-7}$	$9.1 \cdot 10^{-5}$	9/1	0.0002	0.0715	9/1
20	A_{heur}	ADAM 0.03×100	$6.1 \cdot 10^{-6}$	$1.1 \cdot 10^{-5}$	0/10	0.0060	0.0102	0/10
20	A_{heur}	L-BFGS-B	$1.5 \cdot 10^{-7}$	$2.8 \cdot 10^{-7}$	0/10	0.0002	0.0003	0/10
12	A_{margin}	ADAM 0.03×100	$2.0 \cdot 10^{-4}$	$1.3 \cdot 10^{-3}$	4/6	0.0021	0.0286	4/6
12	A_{margin}	L-BFGS-B	$2.9 \cdot 10^{-4}$	$9.4 \cdot 10^{-4}$	6/4	0.0001	0.0234	6/4
16	A_{margin}	ADAM 0.03×100	$6.1 \cdot 10^{-5}$	$1.5 \cdot 10^{-4}$	1/9	0.0281	0.0834	1/9
16	A_{margin}	L-BFGS-B	$3.2 \cdot 10^{-5}$	$4.8 \cdot 10^{-4}$	3/7	0.0217	0.0792	3/7
20	A_{margin}	ADAM 0.03×100	$1.2 \cdot 10^{-4}$	$3.1 \cdot 10^{-4}$	0/10	0.1060	0.2566	0/10
20	A_{margin}	L-BFGS-B	$1.1 \cdot 10^{-6}$	$5.2 \cdot 10^{-5}$	8/2	0.0010	0.0469	8/2

3 exhausted the gradient budget, and of the eleven that did stop on a tolerance, all eleven stopped on the *objective* criterion and none on the gradient one, with a final $|\nabla|_\infty$ between $2.7 \cdot 10^{-5}$ and $1.2 \cdot 10^{-1}$ over the sixteen. In double precision none of the 16 failed, seven of the sixteen stopped on the gradient criterion, and $|\nabla|_\infty$ ran $1.7 \cdot 10^{-6}$ to $3.4 \cdot 10^{-4}$ — one to three orders of magnitude smaller — using 15–73 gradient evaluations against 27–100. The same block at $N = 20$ (16 runs, on the simulation host) had no failures and no budget exhaustion at either precision, but the same asymmetry: all eight single-precision runs stopped on the objective criterion with $|\nabla|_\infty$ between $5.8 \cdot 10^{-5}$ and $1.9 \cdot 10^{-2}$ after 27–85 gradient evaluations, while four of the eight double-precision runs stopped on the gradient criterion with $|\nabla|_\infty$ between $7.5 \cdot 10^{-7}$ and $1.3 \cdot 10^{-4}$ after 17–27. The claim that survives is therefore “terminated under SciPy’s stopping criteria at single precision”, not “converged”.

What does *not* change is the result. The optimised $\langle H_{\text{norm}} \rangle$ agrees between the two precisions to $6 \cdot 10^{-4}$ in 14 of the 16 paired conditions; the largest disagreement is 0.053, a hundredth of the 3.4–5.5 spread the same instance shows across its five initial angles, and in both of the two largest the single-precision run took more gradient evaluations (62 against 53; the whole budget of 100 against 68) and ends *lower*, not higher, than its double-precision partner. At $N = 20$ the eight pairs agree to $4.7 \cdot 10^{-5}$. A similar expectation does not by itself imply similar mass on the success set — two distributions with the same mean can differ on P_τ — so the diagnostic block also scores each run’s exact final state on what the paper reports (Table S36): over the 24 pairs the two precisions agree on P_F , P_τ , $Q_\tau(1000)$ and D_{cond} to $1.0 \cdot 10^{-5}$, $7 \cdot 10^{-8}$, $3 \cdot 10^{-5}$ and $2 \cdot 10^{-5}$ in the median, and in 22 of the 24 pairs to $3.6 \cdot 10^{-3}$, $1.7 \cdot 10^{-5}$, $4.4 \cdot 10^{-3}$ and $1.0 \cdot 10^{-3}$ at worst; the two exceptions are the two pairs above whose expectations differ by 0.04–0.05, both at A_{margin} with seed 42. At $N = 12$, instance 0, both precisions stopped on the objective criterion (62 against 53 gradient evaluations, $\langle H_{\text{norm}} \rangle -4.317$ against -4.264) and the single-precision state has P_F 0.443 against 0.377, P_τ $7.0 \cdot 10^{-3}$ against $3.1 \cdot 10^{-3}$ and $Q_\tau(1000)$ 0.999 against 0.956; at $N = 16$, instance 1, the single-precision run exhausted the 100-gradient budget while the double-precision run stopped on the objective criterion after 68 ($\langle H_{\text{norm}} \rangle -5.569$ against -5.528), with P_F 0.465

against 0.415, P_τ $1.4 \cdot 10^{-3}$ against $0.7 \cdot 10^{-3}$ and $Q_\tau(1000)$ 0.747 against 0.520. In both pairs the two precisions reached different optimisation outcomes on the same start — the size of the difference is that of a different local minimum, not of rounding — and in both the reading the paper draws does not change: P_τ stays below the uniform-F success probability of the instance ($1.5 \cdot 10^{-2}$ and $1.8 \cdot 10^{-3}$) at either precision. The 1,000-shot batch’s within- τ verdict differs on 8 of the 24 pairs; on every one of them the two states’ P_τ agree to $3.0 \cdot 10^{-6}$ and their $Q_\tau(1000)$ to $6.4 \cdot 10^{-4}$, so the disagreement is the draw (Sec. S-VII). The conclusion of this section rests on those comparisons, not on the stopping reason. Runs now record SciPy’s status, message, iteration count, final gradient norm, tolerances and version, which is what this diagnostic had to be run to recover. The block itself is not portable across hosts at single precision: re-running the $N = 12$ and $N = 16$ runs on a second CPU host reproduced every double-precision row (the same gradient-evaluation counts, $\langle H_{\text{norm}} \rangle$ to $4 \cdot 10^{-14}$), while 14 of the 16 single-precision rows changed their evaluation count and 7 their stopping reason, with $\langle H_{\text{norm}} \rangle$ within $4.2 \cdot 10^{-4}$; the $N = 20$ rows, re-run on the same host, reproduced bit for bit at both precisions. The second-host rows are the `_n12_16_mac.csv` file of the termination diagnostic, beside the `_n12_16.csv` file they are compared with.

The batch columns of the L-BFGS-B arm — the E3 block’s “no feasible shot” and “min-energy shot infeasible” counts, Table S22’s “optimum hit”, and this diagnostic’s within- τ verdict — come from a re-run of that arm: until 2026-09-13 the solver’s L-BFGS-B path handed the sampler the initial angles rather than the optimised ones (Sec. VI-C of the main text), which the recorded states exposed (the batch’s feasible fraction followed `initial_P_F`, $r = 0.99$, not `final_P_F`, $r = -0.30$, over the 300 runs). The optimisation itself was untouched by the fix, and the re-run on the same host reproduced every state column of the 300 + 48 runs bit for bit — $\langle H_{\text{norm}} \rangle$, P_F , P_τ , D_{cond} , the evaluation counts and SciPy’s status and message, at both precisions of the diagnostic (its Q_τ columns differ from the superseded file by at most $2.8 \cdot 10^{-13}$ because the cancellation-free evaluation of Q was adopted after that file was written). Two regression tests pin both optimizer paths: `test_the_sampler_gets_the_final_angles`

inspects the angles the sampler receives (the argument, not the shots), and `test_reproduces_the_state_solve_recorded` re-evaluates the recorded final angles from scratch against the recorded state; the predicted counts of Table S22 compare every batch with its state.

D. Multistart at equal shots

The five E1 seeds of an instance can be re-read as a five-start strategy: optimise five times, draw 1,000 shots from each state, keep the best feasible candidate of the 5,000, with $Q_{multi,i} = 1 - \prod_k (1 - p_{ik})^{1000}$ from the exact one-shot probabilities p_{ik} . The single-start strategy a user can run at the same shot budget draws one seed at random before optimising and samples that state 5,000 times, $Q_{single,i} = (1/5) \sum_k [1 - (1 - p_{ik})^{5000}]$ — the Q of each seed, then the mean, not the mean p put into Q , and not the median seed’s state, which is a summary no one can select in advance. With $a_k = (1 - p_{ik})^{1000}$ the two are $\prod_k a_k$ and $\text{mean}_k(a_k^5)$, so by the AM–GM inequality $Q_{multi,i} \geq Q_{single,i}$ on every instance identically; the increment is the measurement, set against the four extra optimisations it costs. The same pair of strategies is evaluated for reaching the exact optimum. Uniform sampling from the feasible set at 5,000 shots is carried as the no-optimisation control at the same candidate count.

Medians over 30 instances of $Q_\tau(S)$ at $\tau = 1\%$. “1×1k” is one optimisation sampled 1,000 times (the E1 baseline; median seed); “random 1×5k” one optimisation from a randomly drawn seed sampled 5,000 times; “5×1k pooled” five optimisations with 1,000 shots each and all batches pooled. The gain is the per-instance difference of the last two, with its instance-level 95% bootstrap interval and the number of instances on which it exceeds 0.01; “unif-F 5k” is uniform sampling from F at 5,000 shots. The two “Opt” columns ask the same question of the exact optimum, for the random single start and the pooled five starts. Optimisation time is one seed against five.

S-VII. SV1 AGAINST THE LOCAL GPU, AND E4-A IN FULL

The single-run cross-backend, matched-precision, seed-pair and repeat comparisons on Amazon Braket SV1 that Sec. V-D summarises in a sentence (Tables S25–S29, Fig. S12), the full E4-A comparison (Table S30), and the finite-shot bound Sec. V-D applies.

Cross-backend reproducibility: SV1 versus the local GPU. A device-abstraction layer is only worth the name if changing the device changes nothing but the device. We therefore re-ran the Table S2 experiment for $N \in 8, 12, 16, 20$ on AWS Braket SV1 — a different simulator, by a different vendor, at double rather than single precision, reached over the network — selected by nothing but a backend name. Table S25 reports the result.

Every row reads the single-snapshot cache of Sec. IV-D. Table S28 below shows that the SV1 column is one draw, not a measurement: Three runs of each row return three different portfolios at $N \geq 16$. We print one run here because that is

what a benchmark normally prints, and we then measure what that convention costs.

At $N = 8$ and $N = 12$ the two committed artifacts agree to the last bit of a double — not to plotting accuracy, but exactly. At $N = 16$ and $N = 20$ they return different feasible portfolios. The $N = 16$ row is the more instructive of the two: SV1 misses the exact optimum by 0.017%, and the portfolio it returns has a Sharpe ratio 7.7% higher than the optimum’s (1.492 vs. 1.385). The backend that solved the stated problem *worse* produced the better portfolio, which is Sec. V-A’s decoupling arriving by a third route.

The explanation that the cloud simulator is less accurate at larger N does not fit the data. Fig. S12 (right) plots the per-step distance between the two optimizer trajectories:

$|\langle H \rangle_{SV1} - \langle H \rangle_{GPU}|$ is $\approx 1 \times 10^{-6}$ at *every* size and from the first step, and the largest disagreement occurs at $N = 16$ (up to 2.1×10^{-5}), which is also the smallest size at which the answers diverge — no trajectory point out of 50 coincides there. The perturbation does not grow with N ; what grows is the density of near-degenerate solutions that a fixed 1×10^{-6} can separate. This is the same mechanism as the $N = 30$ basin change of Sec. IV-D, reached by a different route: there a 1×10^{-10} change in the *input* moved the answer, here a 1×10^{-6} difference in *floating-point evaluation order* does, and both times the objective moves by a few 1×10^{-4} while the Sharpe ratio moves by tens of percent.

Is it the backend, or is it the precision? The GPU arm of Table S25 runs at single precision and SV1 at double, so a reader can reasonably ask whether a $\approx 1 \times 10^{-6}$ trajectory difference is just fp32 and the backend attribution collapses. We re-ran the GPU arm at double precision to find out.

Reported gap against the enumerated optimum; the last column counts how many distinct portfolios the three arms returned between them.

Matching the precision does not make the answers agree. At $N = 12$ the two double-precision arms *disagree with each other* while the single-precision GPU matches SV1 exactly, and at $N = 20$ all three arms return different portfolios. The higher-precision arm is also the worse one at $N = 12$ — 0.1024% against 0.0000% — which is not a behaviour a precision explanation predicts.

So the disagreement is not fp32. Precision, vendor and evaluation order are three ways of perturbing the same arithmetic at the 1×10^{-6} level, and on this landscape any of them is enough to select a different member of the near-optimal set. The useful statement is not which backend is right but that the *returned solution* is not determined by the formulation at these sizes.

Is it the backend, or is it the sampler? A near-degenerate landscape makes the returned portfolio a *draw* from the near-optimal set, and a benchmark run has more than one unpinned source of randomness: it can differ from another because the optimizer reached different angles, or because the same state was sampled differently. A “shot-RNG control” — the seed sweep re-run with the angles fixed and only the sampling seed changed, which agreed 10 of 10 — was first read as ruling the second out. That control was vacuous: the seed it changed was consulted by no sampler (Sec. VI-C), so the two sweeps

TABLE S24
FIVE STARTS AGAINST ONE, AT EQUAL SHOTS.

N	A	1×1k	random 1×5k	5×1k pooled	Gain [95% CI]	> 0.01 of 30	unif-F 5k	Opt, random	Opt, pooled	Opt. time
12	A_{heur}	0.927	0.905	1.000	+0.095 [+0.095, +0.095]	30	1.000	0.905	1.000	10 → 51 s
12	A_{margin}	1.000	0.945	1.000	+0.055 [+0.026, +0.076]	23	1.000	0.942	1.000	10 → 51 s
16	A_{heur}	0.335	0.684	0.836	+0.153 [+0.152, +0.153]	30	1.000	0.683	0.835	17 → 84 s
16	A_{margin}	0.534	0.639	0.983	+0.336 [+0.316, +0.344]	30	1.000	0.634	0.979	17 → 84 s
20	A_{heur}	0.067	0.305	0.353	+0.046 [+0.015, +0.048]	30	0.873	0.181	0.196	30 → 148 s
20	A_{margin}	0.120	0.463	0.520	+0.060 [+0.036, +0.074]	30	0.873	0.321	0.350	30 → 148 s

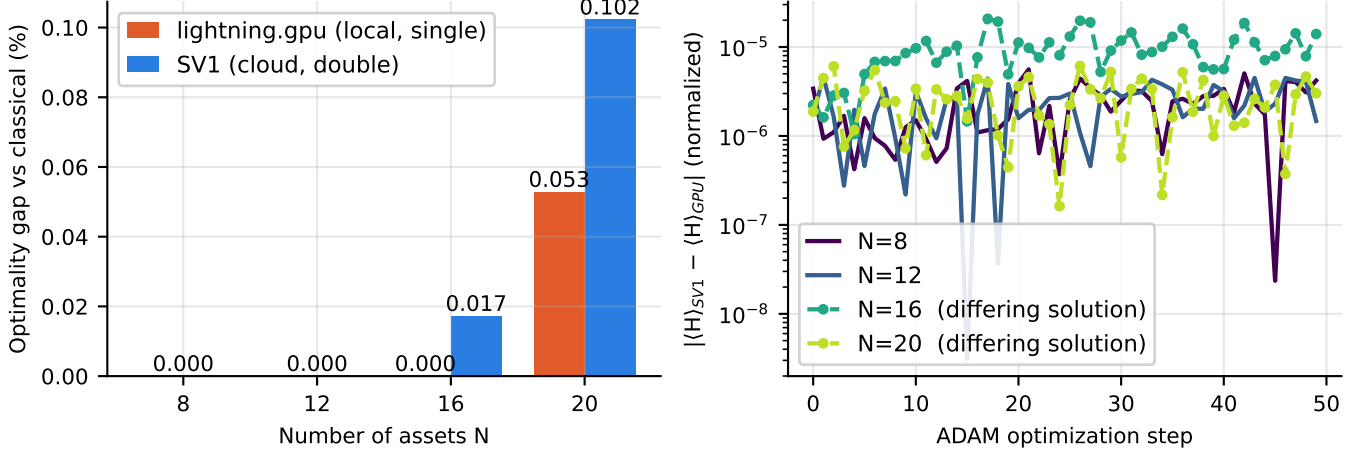


Fig. S12. SV1 vs. lightning.gpu: gap per N and the per-step distance between the two optimizer trajectories (log y).

TABLE S25
SAME CODE, TWO BACKENDS (P = 2, SEED 42, 50 ADAM STEPS, 1000 SHOTS).

N	E, lightning.gpu	E, SV1	Bitwise equal	Gap, GPU	Gap, SV1	Sharpe, GPU	Sharpe, SV1	t, GPU	t, SV1
8	−36.2526116808	−36.2526116808	✓	0.000%	0.000%	1.322	1.322	5.3 s	172 s
12	−52.2526116807	−52.2526116807	✓	0.000%	0.000%	1.322	1.322	10 s	179 s
16	−153.2671105966	−153.2406894624	×	0.000%	0.017%	1.385	1.492	17 s	257 s
20	−335.9747295240	−335.8084974079	×	0.053%	0.102%	1.532	1.222	29 s	858 s

TABLE S26
THE SAME EXPERIMENT AT MATCHED PRECISION.

N	GPU, single	GPU, double	SV1, double	Distinct answers
8	0.0000%	0.0000%	0.0000%	1
12	0.0000%	0.1024%	0.0000%	2
16	0.0000%	0.0000%	0.0172%	2
20	0.0528%	0.0938%	0.1023%	3

re-drew identical shots and agreed because the samples were the same. The separation is made properly below, with the angles fixed and ten genuinely independent batches on each backend (E4-A), after the remaining single-run evidence.

Same instance, same seed, same configuration on each side; only the backend differs.

The one agreement is the reference seed, 42 — and it disagrees with Table S25, which reports SV1 missing the optimum at N = 16 for what is nominally the same configuration. That is not a contradiction between the tables; it is the same finding twice. The two SV1 columns are separate submissions, and Table S28 shows what that costs: three runs of one configuration return three different portfolios at this size. Table S25 prints the submission made for it, Table S27

the one made for the seed sweep, and they differ by 2.6×10^{-2} in energy. A reader comparing the two rows is looking at the paper’s own claim about the cloud simulator, not at an error.

The row worth dwelling on is N = 16, seed 43: the run of Sec. V-B that returned no feasible shot on the GPU (11.22% by its lowest-energy shot) does not happen on SV1, which reports 0.17% for the same configuration. A single-backend seed sweep would have found either the tail or its absence depending on where it ran.

Ruling that out raised a sharper question, because SV1 also disagrees with *itself*. We therefore ran the identical configuration three times at each size — same seed, same depth, same optimizer budget, same shot count, same committed price cache, nothing varied but the submission — and counted

TABLE S27
THE TEN CROSS-BACKEND PAIRS BEHIND “1 OF 10”.

N	seed	Same portfolio	Gap, SV1	Gap, GPU
16	42	yes	0.0000%	0.0000%
16	43	no	0.1703%	11.2246%
16	44	no	0.0000%	0.0041%
16	45	no	0.0172%	0.0041%
16	46	no	0.0270%	0.0230%
20	42	no	0.0860%	0.0528%
20	43	no	0.0526%	0.0979%
20	44	no	0.0177%	0.0089%
20	45	no	0.0712%	0.0049%
20	46	no	0.0035%	0.0220%

how many distinct portfolios came back.

Nothing is varied between runs. The GPU column is the same configuration on `lightning.gpu`.

Every run at $N \geq 16$ returned a different portfolio. At $N = 20$ the four runs in that row — three on SV1, one on the GPU — produced four distinct answers, spanning Sharpe ratios from 1.026 to 1.532. And the small sizes are the control that makes this a mechanism rather than an anecdote: at $N = 8$ and $N = 12$ the three cloud runs are bitwise identical to each other *and* to the GPU, so the cloud is not simply unstable. What changes between $N = 12$ and $N = 16$ is not the service, it is $C(N, K)$ — 66 feasible portfolios against 560 — and with them the density of solutions a floating-point perturbation can separate.

The convergence traces show the optimizer trajectories differ: at $N = 20$ across 50 recorded steps 2 agree and 48 do not, with a maximum difference of 5.2×10^{-15} and a disagreement already present at step

- 1) At $N = 16$ the two runs differ by 4.4×10^{-16} at step 0 — one unit in the last place of a double — and

end in portfolios whose Sharpe ratios differ by 7.7%. SV1’s analytic evaluation is therefore not bit-reproducible between runs. Observing the ULP difference and the different final portfolio in the same pair of runs does not by itself show that the one caused the other: SV1’s final sampling is a second unpinned source, and we have not re-sampled a fixed state on SV1 to separate them (Sec. VII).

That completes a hierarchy worth stating plainly, because it is not the one a user would assume (Table S29).

The managed cloud simulator is the least reproducible of the three we tested, and the only one whose randomness cannot be pinned from the client. A cross-backend check therefore measures backend *and* run, and 5.2×10^{-15} of run-to-run drift in the trajectory accompanies a changed answer on this landscape — the same sensitivity Sec. IV-D measures at 4×10^{-10} in the input. The fixed-angle experiment that follows shows the sampler alone changes the answer as often.

A. E4-A in full

Left: distance between each arm’s exact output probabilities and the GPU double-precision ones at the same angles, maximum over the twelve circuits — the largest per-state difference on F , total variation on $F \cup \perp$ (one outcome per feasible bitstring and one symbol for “no feasible shot”, $p_{\perp} = 1 - P_F$), total variation on F alone (half the L^1 distance of

two unnormalised restrictions, for reference), and the relative change in P_{τ} . These are probabilities, not quantum states. Right: distinct best-feasible portfolios over ten 1,000-shot batches of one state, range over the four circuits of each size, the Monte Carlo 95% interval under i.i.d. draws from the arm’s own distribution in the last row; batches with no feasible shot at $N = 16$, seed 43, in the last column.

B. The finite-shot bound on the selected output

Let P and P' be two one-shot output distributions on the same outcome set with total variation distance $d = \text{TV}(P, P')$, and let a batch be S independent shots to which a deterministic selection rule (here: the lowest- E_A feasible shot, or “no output”) is applied. By the coupling characterisation of total variation there is a joint distribution of one shot from P and one from P' under which the two shots differ with probability exactly d ; drawing the S shots of both batches jointly under S independent copies of that coupling, the two batches are identical with probability $(1 - d)^S$, and identical batches give identical selected outputs. Hence

$$\begin{aligned} \text{TV}(\text{selected output under } P, \text{ selected output under } P') \\ \leq 1 - (1 - d)^S \leq Sd. \end{aligned}$$

The bound is conservative — it counts any differing shot as a differing output — and it needs no assumption about the structure of F . With the maxima of Table S30 at $S = 1,000$ it gives $\leq 4.3 \cdot 10^{-12}$ for the single-threaded CPU, $\leq 2.6 \cdot 10^{-11}$ for SV1 and $\leq 6.1 \cdot 10^{-4}$ for the single-precision arm: these are bounds on how far the arithmetic can move the *distribution* of the selected portfolio at fixed angles. They are not bounds on how often two independently sampled batches return different portfolios — at $S = 1$ two identical fair coins have $d = 0$ and disagree on half of independent draws — which is why ten independent batches of one state can return nine or ten distinct portfolios at $N = 20$ while the distributions they were drawn from are within 10^{-14} of each other. `tests/test_e4_bounds.py` checks the inequality numerically and keeps the two readings apart. This is a derivation from the paper’s own definitions, not a new measurement, and it is what makes E4-B (optimising on SV1 to compare final angles) unnecessary for the claim Sec. V-D makes; E4-B would be needed only to attribute the disagreement of the single runs to backend-dependent optimisation trajectories, which the paper does not do.

TABLE S28
THREE SV1 RUNS OF ONE CONFIGURATION (N, SEED 42, P = 2, 50 STEPS, 1000 SHOTS, DOUBLE).

N	C(N,K)	Distinct answers in 3 SV1 runs	Sharpe range across runs	GPU
8	28	1 (bitwise identical)	1.3219 (no spread)	1.3219
12	66	1 (bitwise identical)	1.3219 (no spread)	1.3219
16	560	3	1.3768 – 1.4920 (8.4%)	1.3851
20	4,845	3	1.0256 – 1.2608 (22.9%)	1.5323

TABLE S29
RUN-TO-RUN REPRODUCIBILITY OF THE THREE EXECUTION ENVIRONMENTS.

Environment	Reproducible?	Evidence
lightning.gpu (local)	yes, <i>bitwise</i>	3 of 3 runs identical
lightning.qubit (local)	only if pinned	bitwise at OMP_NUM_THREADS=1; $\sim 1 \times 10^{-13}$ drift otherwise
SV1 (managed cloud)	no	48 of 50 trace points differ; 3 of 3 runs give different answers at $N \geq 16$

TABLE S30
FIXED ANGLES ON FOUR BACKENDS (E4-A), FULL COMPARISON.

Arm	max $ \Delta p $ on F	TV on $F \cup \perp$	TV on F	$ \Delta P_\tau / P_\tau $	Distinct best, N = 12 (seed 42 / 43)	N = 16 (42 / 43)	N = 20	No-feasible-shot batches, N = 16 seed 43
GPU (reference)	double —	—	—	—	1–2 / 5–6	3–4 / 1–5	10	5–9
CPU double, 1 thread	$2.7 \cdot 10^{-17}$	$4.3 \cdot 10^{-15}$	$2.1 \cdot 10^{-15}$	$2.1 \cdot 10^{-14}$	1–2 / 6–10	3–5 / 3	10	7
GPU single	$5.0 \cdot 10^{-9}$	$6.1 \cdot 10^{-7}$	$3.0 \cdot 10^{-7}$	$5.5 \cdot 10^{-6}$	1 / 6–7	3–4 / 1	10	9
SV1 double (cloud)	$5.3 \cdot 10^{-16}$	$2.6 \cdot 10^{-14}$	$1.3 \cdot 10^{-14}$	$2.8 \cdot 10^{-13}$	2 / 7–8	3–4 / 3–4	9–10	6–7
i.i.d. from own distribution (MC 95%)					1–3 / 5–9	2–6 / 1–7.5	9–10	—

S-VIII. IONQ FORTE-1: THE FOUR TASKS, THE FIDELITY MODEL AND TWO DEFECTS

Four tasks on IonQ Forte-1 (us-east-1), \$33.20 total. Each is the configuration of Table S2 — $p = 2$, 50 ADAM steps — with the optimizer on the local GPU and a single 100-shot sampling circuit on hardware. That split is not a convenience: a QPU cannot use adjoint differentiation, so an optimization loop submitted there becomes one billable task per parameter-shift circuit — $1 + 2 \cdot p \cdot [C(N, 2) + 2N]$ per evaluation, 177 at $N = 8$ and 921 at $N = 20$ for $p = 2$ (Sec. VI-B) — and the optimizer was never submitted to hardware.

The fidelity model. Forte-1 reports a two-qubit fidelity of 0.9905 and a native ZZ gate, so each of the QUBO’s $C(N, 2)$ quadratic terms is one two-qubit gate per QAOA layer and the dense mean–variance QUBO needs $2 \cdot C(N, 2)$ entangling gates at $p = 2$ — 56, 132, 240 and 380 at $N = 8, 12, 16$ and 20. Multiplying the published per-gate fidelity over that count (with SPAM at 0.9938 per qubit) gives the predicted circuit fidelities of Table S31 — a prediction from a simplified product model, not a measured state, process or output-distribution fidelity, and none was measured. The cardinality penalty makes the interaction graph complete regardless of the covariance structure, so the gate count is a property of the encoding; [11] reaches the same point from the annealing side via chain-break fractions of 83–92%. The model did not predict usefulness: at 9% predicted fidelity the $N = 16$ task returned 18% feasible shots, a higher feasible fraction than at $N = 12$, and a portfolio 0.0438% from the exact optimum. What the four tasks establish is narrow: at $N = 16$ the prescribed circuit executed, 18 of its 100 shots were

TABLE S31
PREDICTED CIRCUIT FIDELITY ON IONQ FORTE-1 AT $P = 2$, FROM ITS PUBLISHED CALIBRATION.

N	2-qubit gates	Predicted fidelity
8	56	56%
12	132	26%
16	240	9%
20	380	2%

feasible, and the best of them was 0.0438% from the exact optimum. Whether that is typical of $N = 16$, and where the usable range ends, four tasks cannot say — the two $N = 8$ runs differ by more between themselves (25% and 7% feasible, intervals [17.5, 34.3] and [3.4, 13.7]) than $N = 8$ differs from $N = 16$.

Feasible-shot counts carry a Wilson 95% interval for 100 independent draws; the counts themselves are exact (see the provenance note below), the interval is the sampling uncertainty in reading them as a rate.

Where these integers come from. Forte-1 does not return a shot sequence. Each task’s result is the nested `program_set_task_result` schema, whose executable record holds `measurementProbabilities` — a map from bitstring to frequency — and no `measurements` array. Every probability in all four tasks is an exact multiple of $1/100$ and each map sums to exactly 1, against `requestedShots = successfulShots = 100` and `deviceParameters = None` in the task metadata, so the counts in this table are the empirical frequencies of 100 shots recovered without loss, not estimates and not post-processed by any error-mitigation

TABLE S32
IONQ FORTE-1, 100 SHOTS PER RUN.

N	seed	Distinct strings	Feasible shots [95% CI]	Noiseless feasible	Retention	Gap	Sharpe vs exact optimum
8	42	60	25% [17.5, 34.3]	40.2%	0.62	0.2809%	1.3380 vs 1.3219 (+1.2%)
8	43	79	7% [3.4, 13.7]	9.9%	0.71	0.3917%	1.0889 vs 1.3219 (−17.6%)
12	42	92	11% [6.3, 18.6]	16.4%	0.67	0.0000%	1.3219 vs 1.3219 (exact)
16	42	95	18% [11.7, 26.7]	39.2%	0.46	0.0438%	1.2836 vs 1.3851 (−7.3%)

setting we requested. What is not recoverable is the *order* of the shots, which nothing here uses. The four task records are committed with the paper, so this chain can be re-walked; Sec. VI-C reports what the client library did with the same records before they were read this way.

Three observations follow, and none of them is a hardware limit. Hardware retained 0.46–0.71 of the noiseless feasible fraction across the four runs, with the lowest value at the largest size; the noiseless fraction itself ranges from 40.2% to 9.9% between two seeds at the *same* size, so over these four tasks the initial angles moved the feasible fraction more than the device did — an observation about four tasks at one calibration, which cannot rank noise against initialisation in general. At $N = 12$, 89 of 100 shots violated the cardinality constraint and the remaining 11 contained the exact optimum, so a benchmark reporting only the best shot would record a 0.0000% gap from a circuit whose product-model fidelity is 26%; the quality of the best feasible outcome and the frequency of feasible outcomes are different quantities and are reported apart. And the two $N = 8$ runs report gaps 0.11 percentage points apart (0.2809% and 0.3917%) and Sharpe ratios 19 percentage points apart, one above the exact optimum’s and one 17.6% below it — same device, same instance, same shot count; only the initial angles differ.

Two defects worth reporting, and the second is the first one’s fix. The first three hardware runs returned an energy of exactly 0.0 at a 100% gap. The device returns its shots under a nested result schema that the client library we used does not read and does not raise on; it returned an array that was not the measurements, while the real shots sat in object storage the whole time. A framework that turns an \$8.30 hardware result into a plausible-looking zero is a reproducibility hazard of a different kind from those in Sec. IV-D: not an unstable input, but a silent misread of a correct output. We now read hardware shots from the task record rather than through the library.

That fix then failed on its first contact with hardware, for a reason worth more than the API detail behind it. It located the completed task through a handle the library sets on one execution path and not on the path hardware sampling actually takes. The $N = 16$ run submitted its task, the task completed, \$8.30 was billed, and the run died reading a handle that was never set — the code written to stop losing hardware results lost one, because it had only ever been exercised against a simulator. **A recovery path reachable only by spending money will not be tested, and will therefore be wrong when it is needed.** The current version asks the service which task this run created, requires the answer to be exactly one, and refuses to guess otherwise; a wrong guess would

attribute another run’s shots to this row, which is worse than an exception because the number would look plausible. Five tests now cover it against a stubbed client, at no cost. Both defects, and the code, are in the repository.

S-IX. REPRODUCIBILITY NOTES, THE DEFECTS THAT CHANGED RESULTS, AND RESOURCES

A. The four input-reproducibility rules, and what each cost

All conditions are in code, seeds are named and the window is pinned. Four prerequisites were learned by losing results to them, and are stated because each is cheap to satisfy and easy to miss. (i) *Pin values, not dates.* `yfinance` revises adjusted closes between calls, so two runs on the identical date range produced QUBOs differing at the 10^{-10} level and single-precision QAOA occasionally amplified that into a different local minimum. The raw close series are cached to disk at full precision and read back with round-trip parsing (two fifty-ticker snapshots, ≈ 1.2 MB); the released package carries the derived μ and Σ of each window rather than the third-party series, and every instance’s inputs are a bit-exact slice of them, so every number reproduces from a clone with no network access. Re-running the single-instance benchmark on the cache moved fourteen of its fifteen rows by $\sim 10^{-7}$ (the same assets) and the $N = 30$ QAOA row into a different near-degenerate basin — $5 \cdot 10^{-5}$ in objective, 9% in Sharpe ratio — under an input perturbation four orders of magnitude below any reported digit. (ii) *One snapshot, not one per experiment.* A per-size cache held a sixteen-asset file fetched seven minutes after the others and revised by up to $4.2 \cdot 10^{-7}$; every instance now slices one fifty-ticker snapshot. Re-deriving that size turned the defect into a controlled measurement: under a $4 \cdot 10^{-10}$ relative change in the QUBO, 9 of 60 grid runs changed solution and the worst gap moved from 0.49% to 11.07%. (iii) *Canonical memory layout.* `DataFrame.cov()` sums in layout order, so identical prices gave last-bit-different covariances on two code paths; statistics are computed on a C-contiguous float64 copy. (iv) *Thread count.* `lightning.qubit` reduces in thread order and drifts by $\sim 10^{-13}$ between runs unless `OMP_NUM_THREADS=1`; `lightning.gpu` is bitwise reproducible regardless. The experiment scripts pin the thread count before importing a simulator and a test asserts it. Figures are regenerated from the committed CSVs with embedded timestamps suppressed, so a changed figure file means changed data.

B. Implementation defects that changed results, in the order found

Seven implementation defects changed reported numbers; six were found before any number in Sec. V of the main

text was produced, the seventh after. Sec. VI-C of the main text states the four correspondences they left as checks; this is the record of the defects themselves, in the order found. *Before Sec. V.* (1) The angles were a plain array rather than a differentiable tensor, so the optimizer returned them unchanged — a flat “convergence” curve that would have been reported as optimized QAOA. (2) The Braket device rejects computational-basis `sample(wires=...)`; PauliZ is sampled per wire and mapped to bits. (3) The expectation varies in γ at frequencies set by the differences between H_C ’s eigenvalues, which grow with the coefficients (and with their number); at $N = 12$ the expectation varied faster in γ than one ADAM step of 0.1 moved it, and the cost Hamiltonian is therefore divided by its largest coefficient (Sec. III-A) — a choice of parameterisation, not a scale law for the spectrum. At A_{heur} the objective’s whole feasible range is 1.5–1.6% of that coefficient s_H in the median (Sec. V-B), so the normalised landscape is almost entirely the constraint, which is the context for the near-uniform conditional distribution of Sec. V-B. (4), (5) On the hardware path, a nested result schema the client library misread into an energy of exactly 0.0, and a recovery path that had only ever been exercised against a simulator (Sec. S-VIII). (6) While building E4-A, a “shot-RNG control” had re-seeded NumPy’s global generator before the sampling circuit, but the Lightning devices draw a seed from that generator once, at construction, and sample from a private generator thereafter, so the re-seeding changed nothing and two sweeps reported as varying the shots re-drew identical ones. A control that cannot fail is not a control, and a reproducibility claim about a sampler has to name the generator it seeds; E4-A seeds the device’s own.

After Sec. V. (7) An integration error in the L-BFGS-B wrapper left the final sampling call referencing the initial parameter vector, whereas the analytic state quantities were evaluated at the returned final parameters: the ADAM loop rebinds the variable the sampler reads to the array each step returns, but the L-BFGS-B wrapper received a copy of that variable and returned its result in another. The batch columns of the E3 L-BFGS-B arm (“no feasible shot” and “min-energy shot infeasible” in Table II, “optimum hit” in Table VI) therefore described unoptimised states while its state columns ($\langle H_{\text{norm}} \rangle$, P_F , P_τ , Q_τ) did not. It was caught because every run records its exact final state beside its batch: the batch’s feasible fraction tracked the initial state’s P_F ($r = 0.99$) and not the final state’s ($r = -0.30$), and the batch counts fell outside what the recorded states predict under independent draws (Table S22) — a comparison the unaffected ADAM arms pass. The handoff was corrected, regression tests check on both optimizer paths that state evaluation and sampling receive the same final parameters, and the affected finite-shot records were regenerated by re-running the arm with the fix on the same host: every recorded state of the 300 runs — $\langle H_{\text{norm}} \rangle$, P_F , P_τ , the initial state and the evaluation counts — reproduced bit for bit, the batch columns printed are the re-run’s (the batch’s feasible fraction now tracks the final state’s P_F , $r = 0.998$), and the re-run’s counts sit within 3.0 hits and 2.7 no-feasible runs of the independent-draw prediction in every cell, as the ADAM cells do.

C. Resources in full

GPU time is wall-clock on the host of Sec. IV-B; dollars are AWS list price as billed.

The Braket total is \$37.90 against a \$50 budget; the account carries no idle charge, so the deployed footprint is retained.

S-X. SUPPORTING RE-ANALYSES

Five re-aggregations of data the paper already held, each answering one question the main text leaves open, and one evidence table. None re-ran a QAOA optimisation except the precision diagnostic of Sec. S-VI, whose optimised values no claim depends on.

A. Initial $\rightarrow$ final at a fixed penalty weight

Sec. V-B separates what optimisation does at a fixed A from what changing A does after optimisation. For every completed E1 run the exact initial and final states are paired within the run; the difference is reduced to one value per instance by the median over the five seeds and summarised over the thirty instances by its median and bootstrap interval (Sec. IV-E). R_τ is q_τ / u_τ from P_τ / P_F against the instance’s share $|F_\tau|/|F|$; D_{cond} is from `paper01_e1_conditional_tv.csv`. From `paper01_e1_initial_final.csv`.

Medians over instances of the seed-median initial and final levels, the paired median difference with its 95% interval, and the count of instances on which the quantity increased.

B. The objective’s scale in the normalised Hamiltonian

Sec. V-B and Sec. VI-C give the scale of the objective against the coefficient that normalises the Hamiltonian at A_{heur} ; three different numbers could be meant by that. The largest entry of the QUBO matrix and the scale $s_H = \max(\max|h_i|, \max|J_{ij}|)$ that normalises the circuit’s Hamiltonian are different numbers, since each h_i sums several entries of Q, and the quantity the argument needs is neither: it is $\epsilon_F(A) = \Delta_F / s_H(A)$, the objective’s whole range over the feasible set in units of the Hamiltonian the optimiser descends (on F the penalty vanishes and the constant cancels in differences). `coefficient_scale.py` computes all three per instance and setting with no circuit; from `paper01_coefficient_scale_summary.csv`.

Median [min, max] over the thirty instances of each size. “QUBO” is $\max|Q_{\text{objective}}| / \max|Q|$; “Ising” is $s_H(\text{objective only}) / s_H(A)$; ϵ_F is $\Delta_F / s_H(A)$. The 0.5% of the earlier text was the QUBO ratio at $N = 16$.

C. The precision diagnostic, scored on acquisition

Single minus double precision on the same (N, instance, weight, seed) of the diagnostic block (48 runs, 24 pairs), from `paper01_e3_precision_pairs.csv`: each cell is the median (maximum) of the absolute difference over the pairs, and the last column counts the pairs on which the best feasible shot’s within- τ verdict agrees (on each of the 8 disagreeing pairs the two states’ P_τ agree to $3.0 \cdot 10^{-6}$ or better and $Q_\tau(1000)$ to $6.4 \cdot 10^{-4}$, so the disagreement is the draw).

TABLE S33
RESOURCES USED.

Experiment	Runs	Environment	Precision	Time	Tasks	Cost
Single-instance benchmark, $N = 8-30$ (Table S2)	7 QAOA + 8 classical	lightning.gpu	single	3 h 55 min	—	\$0
Seed and depth sweeps, steps $\times p \times$ seed grid (Sec. V-C)	20 + 12 + 180	lightning.gpu	single	4.6 h	—	\$0
Degeneracy census and instance sweep, $N \leq 30$ (Sec. V-A)	30 instances $\times$ 6 sizes; 900 + 15 annealing runs; 90 QAOA	CPU / lightning.gpu	double / single	≈ 1 h	—	\$0
E0 re-scoring (Fig. 1)	354 rows	CPU	double	seconds	—	\$0
E1 cross (Table II)	2,565	lightning.gpu	single	13.9 h	—	\$0
E2 (from E1 states)	0 new	CPU	double	minutes	—	\$0
E3 controls	300 + 300	lightning.gpu	single	3.2 h + 2.4 h	—	\$0
E5 second window (Table S15)	300	lightning.gpu	single	1.65 h	—	\$0
SV1 cross-backend, seed pairs, matched precision, three repeats (Sec. V-D)	4 + 10 + 4 + 12	Braket SV1	double	—	$\approx 1,300$	$\approx \$4.2$ (incl. a \$0.85 pilot that exposed the fallback of Sec. VI-B)
E4-A fixed angles, local arms (Sec. V-D)	3 arms $\times$ 12 circuits $\times$ (1 + 10)	lightning.gpu / lightning.qubit	double, single	3 min	—	\$0
E4-A fixed angles, SV1 (Sec. V-D)	12 circuits $\times$ (1 + 10) + 2 (pilot)	Braket SV1	double	21 min	134	\$0.50
IonQ Forte-1 (Sec. V-D)	4	Forte-1, 100 shots	—	—	4	\$33.20
Deployment and smoke test	—	SV1	—	—	1	\$0.004

TABLE S34
THE SAME RUN BEFORE AND AFTER OPTIMISATION, AT A_{heur} AND A_{margin} .

N	A	Quantity	Initial	Final	Paired diff [95% CI]	Increased of 30
12	A_{heur}	P_F	0.114	0.179	+0.070 [+0.069, +0.070]	30
12	A_{heur}	P_τ	$1.7 \cdot 10^{-3}$	$2.6 \cdot 10^{-3}$	+1.3 $\cdot 10^{-3}$ [+1.2 $\cdot 10^{-3}$, +1.3 $\cdot 10^{-3}$]	30
12	A_{heur}	$Q_\tau(1000)$	0.816	0.927	+0.163 [+0.162, +0.163]	30
12	A_{heur}	E[V]	21.9	10.6	-11.1 [-11.1, -11.1]	0
12	A_{heur}	R_τ	1.001	0.997	+0.004 [+0.002, +0.005]	22
12	A_{heur}	D_{cond}	0.002	0.002	+0.001 [+0.001, +0.001]	28
12	A_{margin}	P_F	0.134	0.357	+0.177 [+0.089, +0.206]	23
12	A_{margin}	P_τ	$2.5 \cdot 10^{-3}$	$7.9 \cdot 10^{-3}$	+6.7 $\cdot 10^{-3}$ [+3.1 $\cdot 10^{-3}$, +7.7 $\cdot 10^{-3}$]	25
12	A_{margin}	$Q_\tau(1000)$	0.921	1.000	+0.055 [+0.010, +0.078]	25
12	A_{margin}	E[V]	16.7	1.5	-11.5 [-11.7, -11.3]	0
12	A_{margin}	R_τ	1.384	1.432	+0.446 [+0.201, +0.543]	26
12	A_{margin}	D_{cond}	0.223	0.175	+0.043 [+0.028, +0.056]	28
16	A_{heur}	P_F	0.035	0.229	+0.128 [+0.128, +0.129]	30
16	A_{heur}	P_τ	$6.3 \cdot 10^{-5}$	$4.1 \cdot 10^{-4}$	+2.3 $\cdot 10^{-4}$ [+2.3 $\cdot 10^{-4}$, +2.3 $\cdot 10^{-4}$]	30
16	A_{heur}	$Q_\tau(1000)$	0.061	0.335	+0.179 [+0.173, +0.182]	30
16	A_{heur}	E[V]	34.8	4.0	-17.2 [-17.2, -17.2]	0
16	A_{heur}	R_τ	1.008	0.997	-0.009 [-0.011, -0.008]	0
16	A_{heur}	D_{cond}	0.002	0.001	-0.001 [-0.001, -0.001]	0
16	A_{margin}	P_F	0.093	0.313	+0.221 [+0.199, +0.279]	30
16	A_{margin}	P_τ	$3.9 \cdot 10^{-4}$	$7.6 \cdot 10^{-4}$	+4.5 $\cdot 10^{-4}$ [+3.8 $\cdot 10^{-4}$, +5.8 $\cdot 10^{-4}$]	28
16	A_{margin}	$Q_\tau(1000)$	0.324	0.534	+0.255 [+0.206, +0.310]	28
16	A_{margin}	E[V]	25.7	1.7	-21.1 [-21.7, -20.2]	0
16	A_{margin}	R_τ	2.033	1.627	-0.225 [-0.628, +0.146]	11
16	A_{margin}	D_{cond}	0.388	0.139	-0.174 [-0.197, -0.132]	2
20	A_{heur}	P_F	0.087	0.167	+0.134 [+0.133, +0.135]	30
20	A_{heur}	P_τ	$3.6 \cdot 10^{-5}$	$6.9 \cdot 10^{-5}$	+5.7 $\cdot 10^{-5}$ [+2.9 $\cdot 10^{-5}$, +5.8 $\cdot 10^{-5}$]	30
20	A_{heur}	$Q_\tau(1000)$	0.035	0.067	+0.054 [+0.028, +0.055]	30
20	A_{heur}	E[V]	45.7	14.3	-18.9 [-18.9, -18.9]	0
20	A_{heur}	R_τ	0.983	1.009	+0.026 [+0.025, +0.031]	29
20	A_{heur}	D_{cond}	0.004	0.002	-0.002 [-0.002, -0.002]	0
20	A_{margin}	P_F	0.048	0.310	+0.242 [+0.234, +0.253]	30
20	A_{margin}	P_τ	$2.8 \cdot 10^{-5}$	$1.3 \cdot 10^{-4}$	+1.0 $\cdot 10^{-4}$ [+9.2 $\cdot 10^{-5}$, +1.5 $\cdot 10^{-4}$]	30
20	A_{margin}	$Q_\tau(1000)$	0.028	0.120	+0.095 [+0.088, +0.138]	30
20	A_{margin}	E[V]	37.0	3.6	-33.3 [-34.4, -32.6]	0
20	A_{margin}	R_τ	1.524	1.012	-0.560 [-0.593, -0.337]	5
20	A_{margin}	D_{cond}	0.145	0.092	-0.045 [-0.062, -0.028]	3

D. The exact-optimum share per E1 setting

The population Fig. 2 draws: every completed E1 run, from `paper01_e1_optimum_hits.csv`. Each cell is hits / runs with a feasible shot (the gap panels of Fig. 2 plot only those) and the share among them; the completed runs per cell are 150, except 45 at $A = 0$ and 105 at the A_{crit} multiples of $N = 12$ (the nine $A_{\text{crit}} = 0$ instances run only $A = 0$, A_{margin} and A_{heur}).

E. Which factor carried the paired P_τ gain

From $P_\tau = P_F \cdot u_\tau \cdot R_\tau$ with u_τ an instance constant, the ratio of the two weights' one-shot success probabilities on the same (instance, seed) factorises exactly, and log turns the product into a sum; the geometric mean over the 150 pairs per size is therefore the decomposition (the medians are levels and do not multiply). No pair has a zero probability on either side. From `paper01_e1_acquisition_factors.csv`.

TABLE S35
OBJECTIVE AGAINST PENALTY, THREE WAYS.

N	A	QUBO ratio	Ising ratio	$\epsilon_F = \Delta_F / s_H$
12	A_{heur}	1.16% [0.66, 1.74]	1.24% [0.72, 2.28]	1.46% [0.79, 2.71]
12	A_{margin}	46.51% [33.52, 66.69]	41.40% [33.05, 52.92]	51.99% [23.98, 84.19]
12	$1.1 \cdot A_{\text{crit}}$	72.52% [47.12, 95.14]	69.06% [47.45, 92.16]	78.04% [30.02, 126.30]
16	A_{heur}	0.54% [0.34, 0.83]	0.88% [0.61, 1.55]	1.51% [0.85, 2.49]
16	A_{margin}	20.41% [12.97, 37.22]	28.68% [22.72, 35.25]	45.48% [35.14, 75.81]
16	$1.1 \cdot A_{\text{crit}}$	50.69% [36.10, 89.05]	52.26% [42.87, 83.21]	86.85% [53.54, 178.95]
20	A_{heur}	0.34% [0.19, 0.49]	0.72% [0.51, 1.24]	1.63% [1.05, 2.50]
20	A_{margin}	10.17% [7.39, 15.97]	21.73% [17.62, 25.94]	47.30% [34.45, 67.34]
20	$1.1 \cdot A_{\text{crit}}$	28.61% [17.71, 69.87]	43.43% [35.67, 70.49]	97.56% [64.38, 198.26]

TABLE S36
THE PRECISION DIAGNOSTIC SCORED ON ACQUISITION.

N	Pairs	$ \Delta \langle H_{\text{norm}} \rangle $	$ \Delta P_F $	$ \Delta P_\tau $	$ \Delta Q_\tau(1000) $	$ \Delta D_{\text{cond}} $	Verdict agrees
12	8	$2.8 \cdot 10^{-6}$ ($5.3 \cdot 10^{-2}$)	$1.2 \cdot 10^{-5}$ ($6.6 \cdot 10^{-2}$)	$2.0 \cdot 10^{-7}$ ($3.8 \cdot 10^{-3}$)	$1.4 \cdot 10^{-4}$ ($4.3 \cdot 10^{-2}$)	$1.9 \cdot 10^{-5}$ ($9.0 \cdot 10^{-2}$)	4
16	8	$1.1 \cdot 10^{-4}$ ($4.1 \cdot 10^{-2}$)	$2.0 \cdot 10^{-5}$ ($5.0 \cdot 10^{-2}$)	$1.1 \cdot 10^{-7}$ ($6.4 \cdot 10^{-4}$)	$1.7 \cdot 10^{-5}$ ($2.3 \cdot 10^{-1}$)	$2.8 \cdot 10^{-5}$ ($8.3 \cdot 10^{-2}$)	5
20	8	$4.2 \cdot 10^{-6}$ ($4.7 \cdot 10^{-5}$)	$1.0 \cdot 10^{-5}$ ($9.4 \cdot 10^{-4}$)	$2.3 \cdot 10^{-9}$ ($5.3 \cdot 10^{-7}$)	$2.3 \cdot 10^{-6}$ ($5.1 \cdot 10^{-4}$)	$7.9 \cdot 10^{-6}$ ($9.8 \cdot 10^{-4}$)	7

TABLE S37
RUNS WHOSE BEST FEASIBLE SHOT IS THE EXACT OPTIMUM, PER SETTING (E1).

N	A = 0	$1.1 \cdot A_{\text{crit}}$	$2 \cdot A_{\text{crit}}$	$5 \cdot A_{\text{crit}}$	$10 \cdot A_{\text{crit}}$	A_{margin}	A_{heur}
12	41/45 (91%)	99/105 (94%)	89/105 (85%)	76/105 (72%)	73/105 (70%)	118/150 (79%)	115/150 (77%)
16	—	89/150 (59%)	72/150 (48%)	59/134 (44%)	54/135 (40%)	56/139 (40%)	51/136 (38%)
20	—	25/150 (17%)	23/150 (15%)	8/150 (5%)	17/150 (11%)	18/150 (12%)	4/149 (3%)

TABLE S38
 $P_\tau(A_{\text{margin}}) / P_\tau(A_{\text{heur}})$ AS P_F RATIO $\times$ R_τ RATIO, ON THE SAME PAIRS.

N	Pairs	P_τ ratio, geometric mean	$= P_F$ ratio $\times$ R_τ ratio	Share of the log gain in P_F	Medians ($P_\tau / P_F / R_\tau$ ratio)	Pairs with R_τ ratio > 1
12	150	1.64	1.35×1.22	60%	1.69 / 1.26 / 1.37	113
16	150	1.59	1.10×1.45	21%	2.41 / 1.26 / 1.56	125
20	150	2.29	1.79×1.28	70%	2.52 / 1.61 / 1.02	88